FARMLINK

Agricultural Marketplace Platform

PHASE I REPORT

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BONAFIDE CERTIFICATE

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"This project addresses the following Sustainable Development Goal(s): **SDG 2 : Zero Hunger and SDG 8 : Decent Work and Economic Growth**"

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ABSTRACT

Smallholder farmers often face difficulties in accessing commercial markets because their individual production quantities may be insufficient to meet the requirements of large buyers. Dependence on multiple intermediaries can further reduce farmer profitability, while commercial buyers require consistent quantity, quality and reliable fulfilment. Additional challenges include the perishable nature of agricultural produce, limited visibility of sales, difficulty in tracking partial sales and complications in calculating payments when produce from multiple farmers is aggregated.

FarmLink – Agricultural Marketplace Platform is a digital agricultural marketplace designed to connect smallholder farmers with commercial buyers through a network of local Collection Centers. The platform introduces a digital layer between farmers, Collection Centers and commercial buyers to improve market connectivity, inventory visibility and operational traceability.

The Collection Center acts as the physical aggregation and control point where produce is received, weighed, inspected, graded, labelled, stored and dispatched. Each accepted quantity is represented as a unique digital batch that remains associated with the supplying farmer, creating a direct relationship between physical produce and its digital record.

FarmLink uses a batch-based inventory system rather than treating all produce as anonymous inventory. Individual batches can also be partially sold, allowing the system to maintain accurate sold and remaining quantities. When a buyer places an order, the system allocates eligible batches using freshness as the primary criterion and farmer rating as a secondary tie-breaker.

The platform also introduces a two-tranche farmer payment model, digital farmer ledgers, quality-based ratings, freshness-based pricing and transparent buyer pricing. The initial implementation follows a full-stack architecture consisting of React with Vite, Tailwind CSS, Node.js with Express, PostgreSQL, Prisma, JWT-based authentication and Git/GitHub.

Keywords: *Agricultural Marketplace, Smallholder Farmers, Collection Center, Digital Batch, Traceability, Inventory Management, Smart Allocation, Farmer Settlement.*

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
API	Application Programming Interface
CC	Collection Center
DFD	Data Flow Diagram
e-NAM	National Agriculture Market (Electronic)
FAO	Food and Agriculture Organization
FPO	Farmer Producer Organization
JWT	JSON Web Token
MVP	Minimum Viable Product
NABARD	National Bank for Agriculture and Rural Development
ORM	Object Relational Mapping
QR	Quick Response (Code)
REST	Representational State Transfer
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 GENERAL

Agricultural supply chains involve the movement of produce from farmers to consumers and commercial buyers through several physical and commercial stages. Smallholder farmers often operate with limited production quantities and may find it difficult to directly access large commercial markets.

A typical agricultural supply chain begins at the farm, where produce is harvested, and ends with the commercial buyer who procures it for onward processing, retail or institutional use. Between these two points the produce is gathered, sorted, transported and stored, and each of these activities introduces handling effort, time delay and additional cost. Because agricultural produce is perishable, the speed and organization of these intermediate stages directly influence the quality of the produce that finally reaches the buyer and, consequently, the value that can be realised for it.

Commercial buyers, on the other hand, require agricultural produce in consistent quantities and grades. A single smallholder farmer may not be capable of satisfying such requirements independently. This creates a need for an organized aggregation mechanism that can combine produce from multiple farmers while maintaining information about individual contributions.

The mismatch between the scale of individual farm production and the scale of commercial demand is the central difficulty that FarmLink attempts to address. Aggregation, in which produce from several farmers is brought to a common point and presented as a single, standardized supply, is a well-understood response to this mismatch. However, aggregation by itself is not sufficient. When produce from many farmers is merged without records, the identity of each contribution is lost, and it becomes difficult to explain to a farmer how much of the produce was sold or what amount is owed. The need is therefore for aggregation that preserves individual contribution.

FarmLink addresses this requirement through a network of local Collection Centers. Farmers supply their produce to Collection Centers, where the produce is

weighed, inspected, graded and recorded. Each accepted quantity is converted into a digital batch, allowing the physical produce to remain associated with its digital record.

The central design principle of FarmLink is traceability. The platform is able to determine who supplied the produce, how much was supplied, where it is currently stored, how much has been sold and how much money is owed to the farmer. This batch-level approach enables the system to maintain accurate inventory, support partial sales and calculate farmer settlements based on actual quantities sold.

The unit through which this traceability is achieved is the digital batch. A batch is created at the Collection Center for every accepted quantity and carries the supplying farmer, the crop, the grade, the original quantity, the remaining quantity, the sold quantity and the relevant timestamps. For example, when a farmer supplies 60 kg of tomatoes, a 60 kg batch is created. If a buyer later purchases 40 kg, the same batch continues to exist with a sold quantity of 40 kg and a remaining quantity of 20 kg. The batch is not discarded or merged with other stock, and the farmer, the Collection Center and the platform can all refer to the same record throughout its life.

1.2 OBJECTIVES

The major objectives of FarmLink are as follows:

1. To connect smallholder farmers with commercial buyers through a digital marketplace.
2. To reduce unnecessary intermediary involvement in agricultural transactions.
3. To provide Collection Centers with digital tools for receiving, weighing, grading and managing produce.
4. To create unique digital batches for traceable agricultural inventory.
5. To maintain inventory at the individual farmer-batch level.
6. To support partial sales of individual produce batches.
7. To automatically allocate buyer orders to suitable farmer batches.
8. To prioritize freshness during batch allocation.

9. To use farmer ratings as a secondary allocation criterion when freshness is equal.
10. To provide transparent farmer payment and ledger management.
11. To support advance and final settlement through a two-tranche payment mechanism.
12. To improve visibility of inventory, orders, sales and farmer earnings.
13. To provide commercial buyers with standardized agricultural inventory.
14. To improve coordination between farmers, Collection Centers and buyers.

The objectives listed above can be grouped into three broad concerns. The first concerns market connectivity: connecting smallholder farmers with commercial buyers, reducing unnecessary intermediary involvement and giving buyers access to standardized inventory through a single marketplace. The second concerns operational control: equipping Collection Centers with digital tools for receiving, weighing, grading and storing produce, creating traceable batches, supporting partial sales and automatically allocating orders with a preference for fresher produce and, where freshness is equal, better-rated farmers.

The third concerns financial and informational transparency: providing a two-tranche payment mechanism, a farmer ledger that separates advances from final settlements, and improved visibility of inventory, orders, sales and earnings for every participant. These concerns are interdependent. Reliable allocation depends on accurate batch records, and trustworthy farmer payments depend on accurate sold quantities, which in turn depend on the same batch records. The objectives are therefore addressed through a single, consistent batch-based data model rather than through independent features.

1.3 EXISTING SYSTEM

In the existing agricultural supply chain, smallholder farmers frequently face difficulty accessing large commercial buyers because their production quantities are relatively small. This reduces their bargaining power and may increase their dependence on intermediaries. The major limitations of the existing system are described below.

Limited Bargaining Power

Individual farmers may not have sufficient quantity to negotiate effectively with large commercial buyers.

Intermediary Dependence

Multiple intermediaries may participate in the supply chain, reducing the farmer's final profit.

Quantity and Quality Requirements

Commercial buyers require consistent quantities and quality grades, which can be difficult for individual farmers to provide.

Perishability

Agricultural produce is perishable and needs to move quickly through the supply chain. Delays can reduce product value.

Fresh produce loses value as it ages. If older stock is allowed to remain unsold while newer stock is dispatched, the older stock approaches spoilage and the loss falls on the farmer who supplied it. Any system that handles aggregated produce must therefore make the age of each quantity visible and use it when deciding what to sell first.

Limited Sales Visibility

Farmers may not have clear information about where their produce was sold or how much of their original batch remains unsold.

Difficult Partial-Sale Tracking

Manual systems may struggle to maintain accurate records when only part of a farmer's batch is sold.

Commercial orders rarely match the quantity supplied by a single farmer. A buyer may need 40 kg from a 60 kg lot, leaving 20 kg that must still be tracked. In paper registers or informal records, the remaining quantity is easily forgotten, double-counted or attributed to the wrong supplier, which weakens both inventory accuracy and payment accuracy.

Payment Complexity

When a buyer purchases aggregated produce from multiple farmers, determining individual farmer payments can become complicated.

Poor Coordination

Collection and transportation activities may not be sufficiently coordinated, resulting in inefficient movement of produce.

FarmLink addresses these issues by introducing a digital layer between farmers, Collection Centers and commercial buyers.

1.4 PROPOSED SYSTEM

FarmLink proposes a three-sided agricultural marketplace consisting of the following participants:

- Farmers
- Collection Centers
- Commercial Buyers

Farmers supply produce to Collection Centers. The Collection Centers function as physical aggregation and inventory nodes where produce is received, weighed, inspected, graded, labelled, stored and dispatched. Commercial buyers use the digital marketplace to find and purchase standardized agricultural inventory.

The basic operational flow of the proposed system is as follows:

Farmer → Collection Center → Weigh + Grade + Record → Create Digital Batch → Store Physical Produce → Marketplace Inventory → Buyer Places Order → Batch Allocation → Collection Center Picks Produce → Dispatch → Delivery Confirmation → Farmer Final Settlement

Unlike a conventional inventory system, FarmLink maintains each farmer's produce as identifiable batches. Therefore, even when produce is aggregated, ownership and quantity information remain traceable. The proposed system also supports partial batch sales, smart inventory allocation, two-tranche farmer payments, digital ledgers, ratings and freshness-based pricing.

CHAPTER 2
LITERATURE SURVEY

2.1 LITERATURE REVIEW

A review of existing literature was carried out to understand the current state of digital agricultural marketplaces, electronic trading platforms and collective farmer participation mechanisms. Table 2.1 summarizes the reviewed works along with the corresponding inference drawn for the design of FarmLink.

Table 2.1: Summary of Literature Survey

S.No.	Author(s) & Year	Focus Area	Inference
1	Das et al. (2024)	e-NAM and Digital Trade	Digital agricultural marketplaces can improve farmer access to wider markets.
2	Argade et al. (2022)	Electronic Marketplaces	Accurate and accessible information is essential for effective digital agricultural transactions.
3	Raghavendra et al. (2023)	Farmer e-Marketing	Digital agricultural systems need to provide accessible and practical workflows for farmers.
4	Agnihotri & Bhattacharya (2021)	Digital Marketplace Usage	Trust and usability are important when developing digital agricultural platforms.
5	Shroff (2021)	Marginal Farmers and Collectives	Aggregation can help overcome the limitations of individual farmer production capacity.
6	Kumar & Kumar (2022)	FPO Business Ecosystem	Digital systems can improve the coordination of collective agricultural activities.
7	Devika et al. (2025)	Agriconnect	Digital platforms can provide useful connections between agricultural producers and markets.
8	Mishra et al. (2025)	Digital E-Marketplaces	Interfaces for agricultural platforms should be simple and accessible.
9	Nandhini et al. (2025)	Digital Marketing	Digital visibility can provide agricultural producers with broader market opportunities.
10	Khan (2025)	Agri E-Commerce	Reliable infrastructure is necessary for digital agricultural commerce.
11	Dave & Sajja (2024)	FPO eAuction	Digital transaction mechanisms can support collective agricultural market participation.

S.No.	Author(s) & Year	Focus Area	Inference
12	Ravi Kumar et al. (2023)	Chilli e-Market Participation	Market information can influence participation in digital agricultural markets.
13	Gummagolmath & Lakshmi (2022)	FPO Federation	Collective aggregation can help overcome small-scale production limitations.
14	Kumar et al. (2023)	FPO Performance	Improved market linkages are important for collective agricultural organizations.
15	Pendyala et al. (2022)	FPO E-Commerce Awareness	Awareness and usability are important for encouraging digital adoption.
16	Singh et al. (2024)	e-Agri Supply Chains	Coordination between supply-chain participants is essential for operational efficiency.
17	Kumar & Parshant (2026)	ARGOMARKET	Web-based agricultural marketplaces can provide a foundation for direct digital transactions.

2.2 COMPARISON AND DISCUSSION

The reviewed works can be grouped according to the aspect of digital agriculture that they address. A first group, comprising the studies on e-NAM and digital trade by Das et al. (2024), the web-based marketplace ARGOMARKET by Kumar and Parshant (2026), Agriconnect by Devika et al. (2025), digital marketing by Nandhini et al. (2025) and agri e-commerce by Khan (2025), concerns the use of digital platforms to connect agricultural producers with wider markets. The inferences drawn from these works indicate that digital visibility and direct digital transactions can broaden market opportunities, and that reliable infrastructure is a prerequisite for agricultural commerce. For FarmLink, they justify the choice of a web-based marketplace as the basis for connecting farmers and commercial buyers.

A second group addresses information, usability and trust. The work of Argade et al. (2022) on electronic marketplaces, Raghavendra et al. (2023) on farmer e-marketing, Agnihotri and Bhattacharya (2021) on digital marketplace usage, Mishra et al. (2025) on digital e-marketplaces, Pendyala et al. (2022) on FPO e-commerce awareness and Ravi Kumar et al. (2023) on participation in a chilli e-market collectively suggest that accurate information, accessible workflows, simple interfaces and awareness influence whether farmers adopt digital agricultural systems. These findings

shaped the role-specific design of FarmLink, in which each participant is shown only the information that is relevant to the tasks that the participant performs.

A third group concerns aggregation and collective participation. Shroff (2021) on marginal farmers and collectives, Kumar and Kumar (2022) on the FPO business ecosystem, Gummagolmath and Lakshmi (2022) on FPO federation, Kumar et al. (2023) on FPO performance and Dave and Sajja (2024) on FPO eAuction indicate that aggregation can help overcome the limitations of individual production capacity, that digital systems can improve coordination of collective activities and that improved market linkages are important for collective organizations. Singh et al. (2024) on e-agri supply chains add that coordination between supply-chain participants is essential for operational efficiency. These works support the use of the Collection Center as an aggregation point and the emphasis on coordination between farmers, Collection Centers and buyers.

Taken together, the reviewed literature establishes that market access, information, usability, aggregation and coordination are each important to digital agricultural markets. As summarized in Table 2.1, however, the reviewed works concentrate on these aspects individually, namely electronic marketplaces, FPO participation, digital marketing, e-auctions or supply-chain coordination. Within the scope of the works reviewed for this project, the management of aggregated produce at the level of the individual farmer's batch, the handling of partially sold batches, the allocation of orders with attention to freshness, and the transparent separation of advance and final payments are not the principal subject of study. This is the gap that FarmLink addresses.

The literature indicates that digital agricultural systems can improve market access, information sharing, collective participation and supply-chain coordination. However, existing approaches generally concentrate on individual aspects such as electronic marketplaces, FPO participation, digital marketing, e-auctions or supply-chain coordination. FarmLink extends this concept by combining physical aggregation with digital traceability, as summarized in Table 2.2.

Table 2.2: Comparison and Discussion – Existing Challenges vs. FarmLink Solutions

Existing Challenge	FarmLink Solution
Small individual quantities	Collection Center aggregation
Intermediary dependence	Direct digital marketplace
Anonymous inventory	Farmer-level digital batches
Difficult partial sales	Divisible batch inventory
Inventory aging	Freshness-first allocation
Equal freshness batches	Rating-based tie-breaker
Complex farmer payments	Two-tranche settlement
Payment visibility issues	Digital farmer ledger
Quality uncertainty	Produce grading
Poor traceability	Batch-level tracking
Limited buyer visibility	Marketplace inventory
Unclear pricing	Transparent cost structure

The primary distinction is that FarmLink does not simply aggregate produce physically. It maintains digital identity and quantity information for each batch throughout the supply chain.

Several of the entries in Table 2.2 are closely connected. Collection Center aggregation answers the problem of small individual quantities, but it would also create anonymous inventory if used alone. Farmer-level digital batches remove this anonymity and, in doing so, make divisible batch inventory possible, since the remaining quantity of a batch can be recorded after a partial sale. Batch-level records then provide the basis for freshness-first allocation, because the harvest timestamp is held for every batch, and for the rating-based tie-breaker, because every batch is linked to a farmer whose rating is known.

2.3 CONCLUSION

The literature survey establishes the importance of digital agricultural marketplaces, collective farmer participation, information sharing and supply-chain coordination. FarmLink builds on these concepts by introducing a Collection Center-based marketplace where agricultural produce remains traceable to individual farmer batches. The system combines aggregation with digital inventory management, order

allocation, payment tracking and traceability. This approach provides a foundation for improving transparency and coordination across farmers, Collection Centers and commercial buyers.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

FarmLink follows a full-stack web architecture. The architecture consists of a React with Vite frontend communicating through a REST API with a Node.js and Express backend, which in turn uses Prisma ORM to interact with a PostgreSQL database. The same infrastructure supports role-based interfaces for Farmers, Collection Centers and Buyers.

The architecture can be viewed as three cooperating layers. The presentation layer is the React application built with Vite and styled with Tailwind CSS; it renders the Farmer, Collection Center and Buyer interfaces and sends user actions to the server as requests. The application layer is the Node.js and Express backend, which exposes REST endpoints, authenticates each request through JWT-based authentication and applies the business rules of the platform, such as batch creation, allocation and settlement calculation. The data layer is the PostgreSQL database, which is accessed exclusively through Prisma ORM and which holds the persistent record of users, farmers, batches, inventory, orders, payments and ratings.

Placing the business rules in the backend rather than in the interface ensures that every role works against the same rules. For example, the remaining quantity of a batch is updated by the server when an order is allocated, and the updated value is what every interface subsequently displays. A change made through one role's interface is therefore immediately consistent with what another role sees.

Farmer

The farmer interacts with the system through a dedicated interface for viewing batches, earnings, ledger information, ratings and upcoming harvest information.

Collection Center

The Collection Center manages physical inwarding, weighing, grading, batch creation, inventory and outward dispatch.

Buyer

The buyer uses the marketplace to search inventory, place orders and monitor fulfilment.

The architecture therefore connects the physical movement of agricultural produce with its corresponding digital records, as illustrated in Figure 3.1.

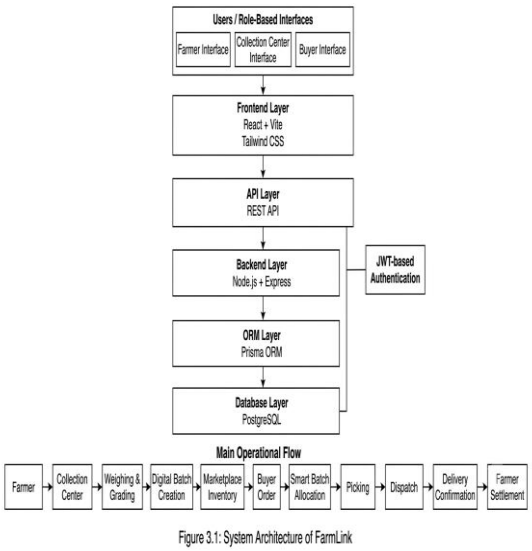


Figure 3.1: System Architecture of FarmLink

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

The FarmLink implementation uses the technology stack summarized in Table 3.1.

Table 3.1: Software Requirements Summary

Component	Technology	Purpose
Frontend	React + Vite	Used to build the web application and user interfaces.
Styling	Tailwind CSS	Used for responsive and maintainable interface development.
Backend	Node.js + Express	Used to implement REST APIs and business logic.
Database	PostgreSQL	Used for persistent storage of users, farmers, batches, inventory, orders, payments and related records.
ORM	Prisma	Used to communicate with PostgreSQL and manage the database schema.

Component	Technology	Purpose
Authentication	JWT-based Authentication	Used to identify users and enforce role-based access.
Version Control	Git + GitHub	Used for source-code management and version tracking.

The software stack summarized in Table 3.1 was chosen so that every component is widely used, well documented and available without licence cost. React with Vite provides a component-based interface in which the dashboards, listing views and order screens can be built from reusable components, and Vite supplies a fast development workflow. Tailwind CSS supports responsive layouts, which is important because farmers are expected to use smartphones while Collection Centers may use computers or tablets.

Node.js with Express allows the backend to be written in the same language as the frontend, which simplifies development for a small team, and Express provides a straightforward means of defining REST routes for each module. PostgreSQL is a relational database, which suits the strongly related records of FarmLink, in which farmers own batches, batches are drawn upon by orders and orders give rise to payments. Prisma ORM defines these relationships in a schema and gives the backend typed access to the database. JWT-based authentication identifies each user and carries the role, which allows the backend to restrict each endpoint to the roles that may use it. Git and GitHub maintain the history of the source code and allow the team members to work on separate parts of the system while keeping their changes organized.

3.2.2 Hardware Requirements

The system is designed to be accessed through standard computing devices without the need for specialized computing infrastructure.

Farmer

- Smartphone
- Internet connection
- Camera/scanning capability for future QR/barcode functionality

Collection Center

- Computer or tablet

- Internet connection
- Barcode/QR scanning capability as the system evolves
- Suitable device for inventory management

Buyer

- Smartphone, tablet or computer
- Internet connection

The camera and scanning capabilities listed for the farmer and for the Collection Center relate to the QR and barcode functionality that is identified as future work. At the current stage of the system, the batch ID and the crate tag generated at inwarding provide the means of associating physical produce with its digital record, and scanning is expected to make this association faster and less error-prone as the system evolves.

3.2.3 Data Requirements

FarmLink requires the following categories of data to support its operations:

Farmer Data

- Farmer identification, profile and contact details
- Rating and performance information

Produce Data

- Crop, quantity and grade
- Harvest timestamp and collection timestamp

Batch Data

- Batch ID, farmer, crop and grade
- Original quantity, remaining quantity and sold quantity
- Storage information

Buyer Data

- Buyer details, required crop and required quantity
- Minimum grade and delivery location

Order Data

- Order ID, product, quantity and allocation

- Delivery information and order status

Financial Data

- Base produce value, advance amount and final settlement
- Pending earnings and cleared earnings

The categories of data listed above are related through the batch. A farmer record is linked to many batches through the supplying farmer, a batch is linked to its crop, grade, quantities and timestamps, and an order is linked to the batches from which it is allocated. Financial data is in turn linked to batches, so that the advance, the final settlement, the pending earnings and the cleared earnings can all be traced to the batch from which they arose. This chain of relationships is what allows a figure shown in the farmer ledger to be explained in terms of produce.

The three quantity fields held for every batch, namely original quantity, remaining quantity and sold quantity, deserve particular mention because they support partial sales. The original quantity is fixed when the batch is created. The sold quantity increases and the remaining quantity decreases as orders are allocated, so that at any instant the sum of the sold and remaining quantities equals the original quantity. Harvest and collection timestamps are held separately, since the harvest timestamp determines the age of the produce used in allocation, whereas the collection timestamp records when the produce entered the Collection Center.

3.2.4 Deployment and Scaling

The FarmLink system is designed so that the initial MVP can be developed incrementally and expanded after the core workflow becomes stable. The system can be scaled by adding more Collection Centers, supporting additional regions, increasing the number of farmers and buyers, supporting additional crops, expanding storage capacity, adding advanced logistics capabilities, supporting additional languages and introducing advanced analytics. The development strategy specifically emphasizes proving the core farmer-to-Collection-Center-to-buyer workflow before introducing more advanced features.

3.3 SYSTEM DESIGN

3.3.1 Activity Diagram

The activity diagram of FarmLink represents the sequential flow of actions performed across the farmer, Collection Center and buyer roles, beginning with the farmer supplying produce and ending with final settlement. Figure 3.2 presents the consolidated activity flow of the system, covering produce inwarding, batch creation, availability checking, smart batch allocation, dispatch and update of farmer inventory following a sale.

Figure 3.2: Activity Diagram of FarmLink

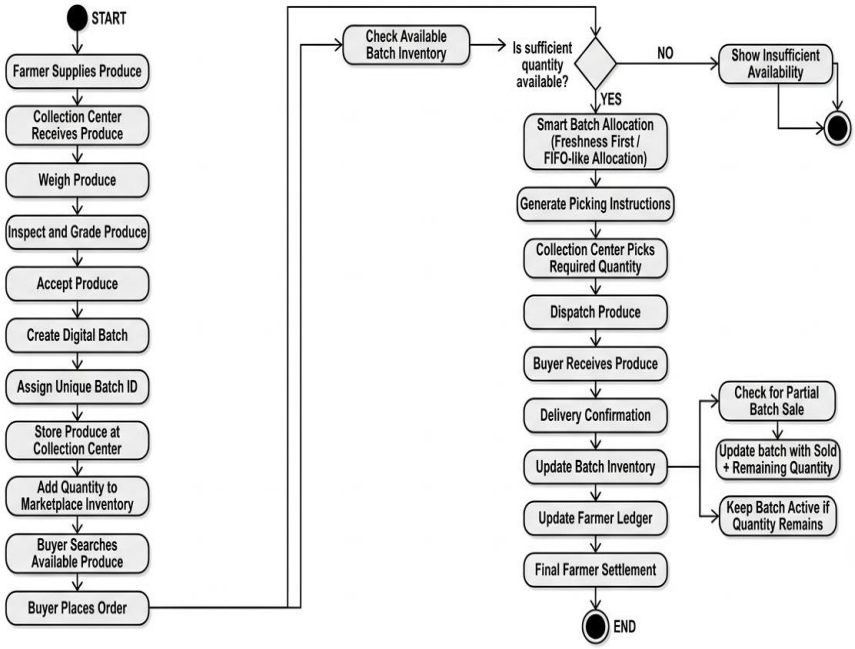


Figure 3.2: Activity Diagram of FarmLink

As shown in Figure 3.2, the flow begins with the farmer supplying produce to the Collection Center, which is weighed, inspected, graded and accepted before a digital batch is created and added to marketplace inventory. When a buyer places an order, the system checks available batch inventory; if sufficient quantity is available, smart batch allocation is performed using a freshness-first, FIFO-like approach, following which picking instructions are generated, the Collection Center dispatches the produce, and the batch inventory and farmer ledger are updated to reflect the sold and remaining quantities.

The activity diagram contains a decision point at the availability check. The system compares the quantity requested by the buyer, for the specified crop and minimum grade, against the remaining quantities of eligible batches. Allocation

proceeds only when the available quantity is sufficient, which prevents the platform from committing produce that does not exist. This check depends directly on the remaining quantity held for each batch, which is why that field is maintained with the same care as the original quantity.

The final activities of the diagram are the update of batch inventory and of the farmer ledger. When an allocation consumes only part of a batch, the sold quantity of that batch is increased and its remaining quantity is reduced, while the batch remains active. When it consumes the whole of a batch, the remaining quantity becomes zero. In both cases the ledger entry for the farmer is updated from the same quantities, so that inventory and finance are changed together and cannot disagree about how much of a batch has been sold.

3.3.2 Data Flow Diagram

The data flow diagram (DFD) illustrates the movement of information between the farmer, Collection Center, commercial buyer and the FarmLink database. Figure 3.3 presents the Level 0 (context-level) DFD, showing the FarmLink Agricultural Marketplace System as a single process exchanging data with its three external entities.

Figure 3.3: Level 0 Data Flow Diagram of FarmLink

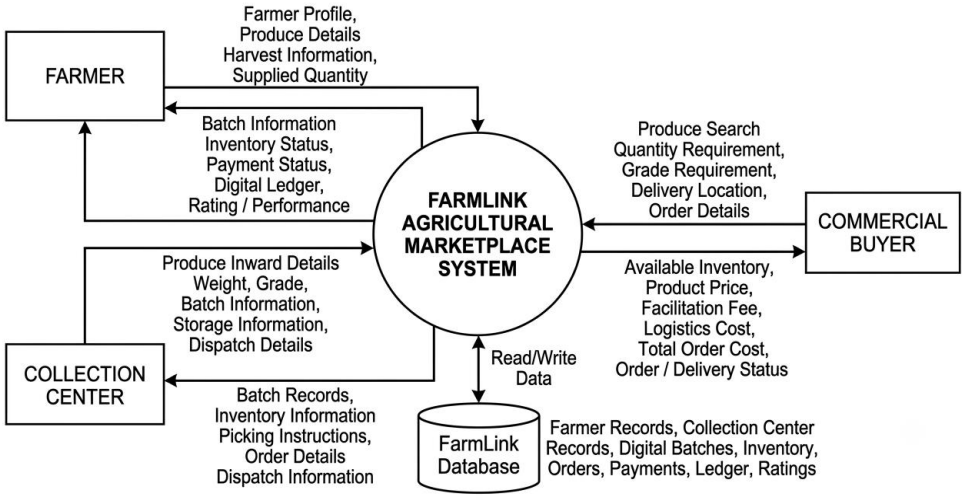


Figure 3.3: Level 0 Data Flow Diagram of FarmLink

Figure 3.4 further decomposes the system into its Level 1 processes: Farmer Management, Collection Center and Batch Management, Buyer and Order Management, Smart Allocation and Fulfilment, and Payment, Ledger and Rating Management, all of which read from and write to the central FarmLink database.

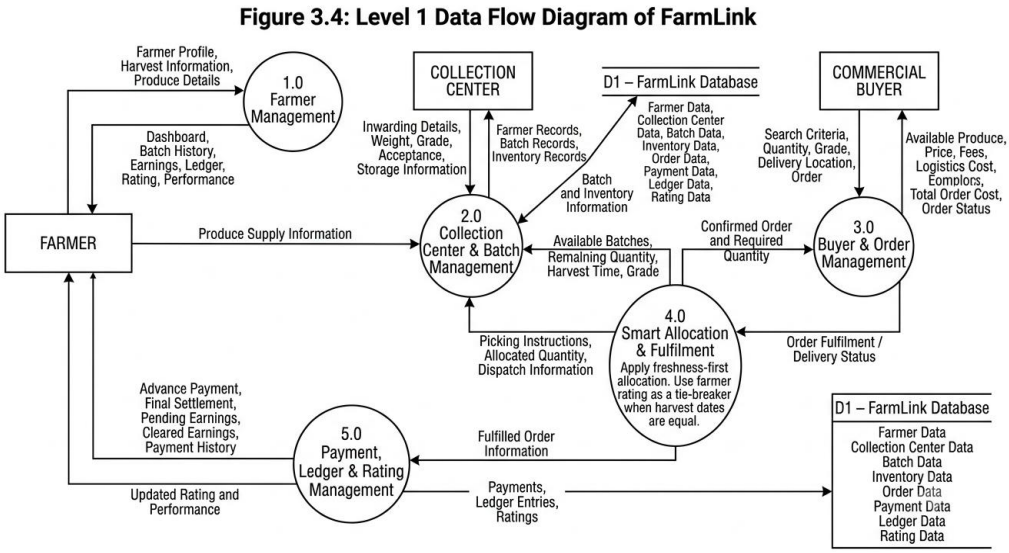


Figure 3.4: Level 1 Data Flow Diagram of FarmLink

Figure 3.4: Level 1 Data Flow Diagram of FarmLink

The five processes of the Level 1 diagram correspond to the modules described in Chapter 4. Farmer Management maintains farmer profiles and presents earnings, ledger and rating information. Collection Center and Batch Management receives inwarding data and maintains batches and inventory. Buyer and Order Management receives search criteria and orders and returns availability and order status. Smart Allocation and Fulfilment reads batches and ratings, selects batches for each order and produces picking instructions. Payment, Ledger and Rating Management calculates advances and final settlements, maintains the ledger and updates ratings from buyer feedback.

Every process reads from and writes to the central FarmLink database, and the processes do not exchange data directly with one another. This is a significant design choice. Because the database is the only shared store, the batch record updated by the allocation process is the same record from which the payment process later calculates a settlement, and no process depends on a private copy of the data.

Together, these diagrams maintain the central traceability relationship of the system:

Farmer → Batch → Inventory → Order Allocation → Sold Quantity → Payment

CHAPTER 4
PROJECT DESCRIPTION

4.1 METHODOLOGY

FarmLink follows a Collection Center-based operational methodology consisting of the following steps.

Step 1 – Farmer Supply

The farmer brings agricultural produce to the Collection Center.

Step 2 – Inwarding

The Collection Center records the farmer, crop, quantity, grade, harvest timestamp and collection timestamp.

Step 3 – Batch Creation

The system generates a unique batch ID. For example, a batch record may appear as follows:

Batch ID: BATCH-TOM-RAMAR-001 | Farmer: Ramar | Crop: Tomato | Grade: A | Quantity: 60 kg

The physical produce can be associated with the digital batch through a QR code or barcode.

Step 4 – Inventory

The digital batch becomes part of Collection Center inventory.

Step 5 – Buyer Order

A commercial buyer searches for produce using crop, required quantity, minimum grade and delivery location.

Step 6 – Batch Allocation

The system identifies suitable batches. Freshness is prioritized first, followed by farmer rating when harvest dates are equal.

Step 7 – Picking and Dispatch

The Collection Center receives digital picking instructions and selects the required batches.

Step 8 – Delivery Confirmation

The buyer's order fulfilment is confirmed.

Step 9 – Final Settlement

The system calculates the farmer's final settlement and updates the farmer ledger.

4.2 MODULES

4.2.1 Farmer Management Module

The Farmer Management Module manages farmer-related information and provides the primary interface for farmers. The farmer dashboard provides information including cleared earnings, pending earnings, total supplied quantity, farmer rating, batch history, financial ledger, performance information and upcoming harvest information. The farmer can view the status of each batch and track the associated financial information.

The module is the farmer's window onto the rest of the system and does not itself alter physical inventory. It reads the batch, ledger and rating records created by the other modules and presents them in a form suited to a farmer who wishes to know what has been supplied, what has been sold and what has been earned. Presenting cleared earnings separately from pending earnings in particular avoids the common confusion in which an advance is mistaken for the final payment for a batch.

The batch history lets the farmer follow each supply from its creation to its current state, for example whether it is wholly unsold, partly sold or fully sold. The rating and performance information shows how the farmer's quality record has developed, and the upcoming harvest information gives the platform and the farmer a shared view of produce that is expected to arrive. Together these items allow the farmer to see the whole relationship with the platform without having to be present at the Collection Center.

4.2.2 Collection Center and Batch Management Module

The Collection Center is the primary physical control point of FarmLink. The module supports produce inwarding, weighing, quality grading, timestamp recording, batch creation, batch identification, inventory updating, crate tagging and outwarding.

The Collection Center can drill down through Crop → Grade → Farmer → Batch → Remaining Quantity → Shelf-Life Status, providing detailed visibility into physical inventory.

The module treats the Collection Center as the point at which physical produce becomes digital inventory. Inwarding captures the quantities and grades; timestamp recording preserves the harvest and collection times that the allocation module later uses; batch identification assigns the unique ID; and crate tagging prepares the produce for identification during storage and picking. Outwarding, at the other end of the lifecycle, records the departure of allocated produce.

The drill-down from Crop to Grade to Farmer to Batch to Remaining Quantity to Shelf-Life Status allows the Collection Center to inspect its inventory at whatever level of detail is needed. A manager may begin with the total available stock of a crop and progress to the individual batches that make it up, noting the remaining quantity and shelf-life status of each. The shelf-life status is particularly useful because it draws attention to batches that are ageing and are therefore likely to be allocated first.

4.2.3 Buyer and Order Management Module

Commercial buyers use the marketplace to search for available agricultural produce using crop, required quantity, minimum grade and delivery location as search criteria. The buyer can view available quantity, estimated delivery, product price, platform facilitation fee, logistics cost and total order cost. This structure provides buyers with greater visibility into the components of the transaction.

4.2.4 Smart Allocation and Fulfilment Module

The allocation module determines which farmer batches should be used to fulfil a buyer order and applies two rules.

Rule 1 – Freshness First

Older produce is allocated first based on harvest time. This follows a FIFO-like approach and helps reduce inventory aging and spoilage.

Rule 2 – Rating as Tie-Breaker

When two batches have the same harvest date, farmer rating is used as the secondary criterion. This provides a balance between product freshness and supplier

performance. After allocation, the Collection Center receives digital picking instructions identifying the required batches or crates.

The tie-breaker is applied only when the freshness rule cannot distinguish between two eligible batches. It never overrides freshness. A highly rated farmer's batch that is newer than another eligible batch is still allocated after the older batch. This preserves the priority given to perishability while allowing quality performance to decide between batches that are otherwise equal.

The allocation proceeds batch by batch. The module sorts the eligible batches by harvest time, applies the rating where harvest dates are equal, and then takes quantity from each batch in turn until the order is satisfied. When the quantity required is smaller than the remaining quantity of a batch, only part of the batch is taken, and the batch continues in inventory with a reduced remaining quantity. When the required quantity exceeds the remaining quantity of a batch, the whole of that remaining quantity is taken and the module moves to the next batch.

After the batches have been selected, the sold and remaining quantities of each batch are updated and the Collection Center receives picking instructions. The instructions identify the batches, and where applicable the crates, from which produce is to be taken. As a result the same allocation decision is reflected in three places: the inventory, the picking instructions and the farmer ledger.

4.2.5 Payment, Ledger and Rating Module

FarmLink uses a two-tranche farmer payment system, consisting of a base advance and a final settlement.

Tranche 1 – Base Advance

After produce is received, weighed, graded and accepted, an initial advance is calculated. For example, for a baseline produce value of ₹10,000, with Tranche 1 set at 70%, the advance paid to the farmer is ₹7,000.

Tranche 2 – Final Settlement

After the corresponding produce is sold and fulfilment is confirmed, the remaining amount becomes eligible for final settlement.

The farmer ledger separates cleared earnings, pending earnings, advances, final settlements, payment history and batch-level transaction history. FarmLink also maintains a dynamic farmer rating based on buyer feedback regarding quality, grade accuracy, freshness and condition upon arrival.

The ledger is a record of financial events at the level of the batch. Each advance, each final settlement and each remaining pending amount is associated with a specific batch, so the farmer can see not only the total earned but also which supplies it came from. Payment history lists these events in order, and the separation of cleared and pending earnings tells the farmer how much has been finally earned and how much remains to be realised.

The rating component completes the module. After an order is delivered, the buyer's feedback on quality, grade accuracy, freshness and condition upon arrival is used to update the rating of the farmers whose batches were part of the order, using a weighted moving average. The rating is displayed in the farmer's dashboard and the marketplace and is used by the allocation module only as a tie-breaker. Quality performance is thereby recognised without permitting the rating to displace the freshness-first rule.

CHAPTER 5
IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

This section presents the implementation outcomes of the major FarmLink modules through screenshots of the working application interface.

5.1.1 Farmer Management

The Farmer Management module is designed to provide farmers with a dedicated dashboard summarizing their supplied quantities, earnings, rating and batch history. The dashboard is intended to consolidate information that would otherwise require the farmer to track produce and payments manually across multiple Collection Center visits.

The Farmer Dashboard displays today's potential earnings, the number of active listings, pending, confirmed and completed orders, and the farmer's current rating, giving the farmer a consolidated view of their engagement with the platform.

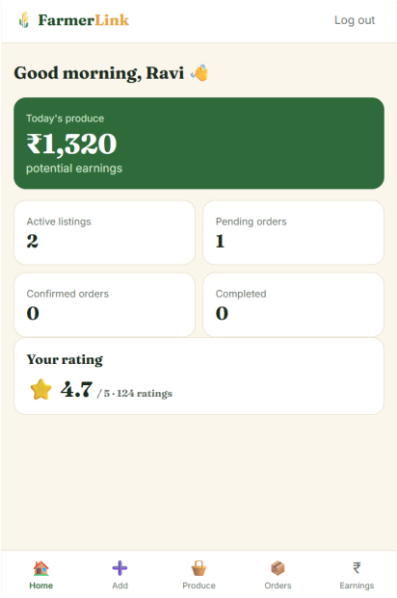


Figure 5.1: Farmer Dashboard

5.1.2 Produce Entry and Inwarding

The Collection Center Inwarding module implements the workflow through which the Collection Center worker records the farmer, crop, weight, grade and harvest

timestamp for incoming produce. The system validates the entered information before a digital batch is created.

The inwarding workflow follows the sequence:

Enter Data → Validate Data → Create Batch → Generate Batch ID → Update Inventory → Calculate Tranche 1 → Update Farmer Ledger → Generate Crate Tag

This workflow establishes the digital identity of the physical produce at the point of entry into the system.

Each step of the inwarding sequence has a specific function. Data entry captures the farmer, crop, weight, grade and harvest timestamp. Validation checks the entries before any record is created, which prevents incomplete batches from entering the inventory. Batch creation and ID generation produce the unique identity of the batch, and the inventory update makes the batch available. The calculation of the first tranche and the ledger update record the advance owed to the farmer at the same moment, so that the physical acceptance of produce and the farmer's financial entitlement are established together. Finally the crate tag is generated so that the physical crates can be identified.

The corresponding interface provides fields for entering the product, category, available quantity, price, harvest date and time, location and grade, along with a description and an option to publish the produce.

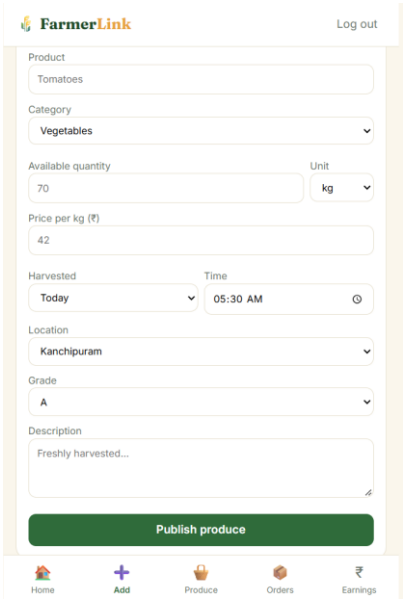


Figure 5.2: Produce Entry Interface

5.1.3 Digital Batch and Inventory Management

The batch system maintains agricultural inventory at the farmer level rather than merging incoming produce into anonymous stock. For example, a Collection Center may simultaneously hold Batch 001 belonging to Ramar (60 kg), Batch 002 belonging to Karthik (40 kg) and Batch 003 belonging to Mani (50 kg). Each quantity remains associated with its corresponding farmer and batch throughout its lifecycle in the system.

The produce inventory view presents each active listing along with its remaining quantity, price, location and harvest age, allowing the farmer to update the available quantity or remove a listing.

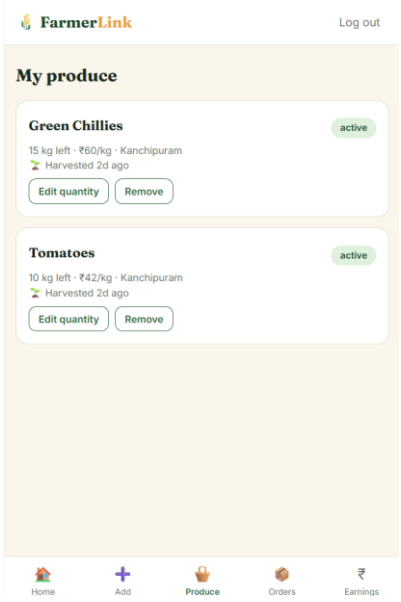


Figure 5.3: Farmer Produce Inventory Management

5.1.4 Partial Batch Sales

FarmLink allows individual batches to be partially sold rather than treating a batch as either fully available or fully sold. For example, if the original batch quantity is 60 kg and a buyer order consumes 40 kg of it, the system records the sold quantity as 40 kg and the remaining quantity as 20 kg, keeping the batch active with its remaining balance instead of incorrectly marking it as completely sold.

The mechanism rests on the three quantity fields held for every batch. Consider the batch of 60 kg described earlier. At creation the original quantity is 60 kg, the sold quantity is 0 kg and the remaining quantity is 60 kg. After a buyer order consumes 40 kg, the original quantity is unchanged at 60 kg, the sold quantity is 40 kg and the remaining quantity is 20 kg. The batch remains active and is listed with 20 kg available. If a later order takes the final 20 kg, the sold quantity becomes 60 kg and the remaining quantity becomes zero, at which point the batch is fully sold.

The significance of this mechanism extends beyond inventory. A commercial order seldom corresponds exactly to the quantity supplied by one farmer, so a system

that can mark a batch only as available or sold would either reject orders that do not match a whole batch or lose track of the unsold portion. By recording the sold and remaining quantities separately, FarmLink allows a single batch to contribute to several orders over time and allows the farmer's ledger to show which part of the batch has been sold and which part is still pending. The sum of the sold and remaining quantities always equals the original quantity, which provides a simple consistency check on the batch record.

5.1.5 Buyer Marketplace

The Buyer Marketplace module allows commercial buyers to search available agricultural produce using keyword search, category, location, sorting and price-range filters. Each listing displays the product, price per kilogram, harvest age, available quantity, grade, supplying farmer and farmer rating, with an option to add the required quantity to the cart.

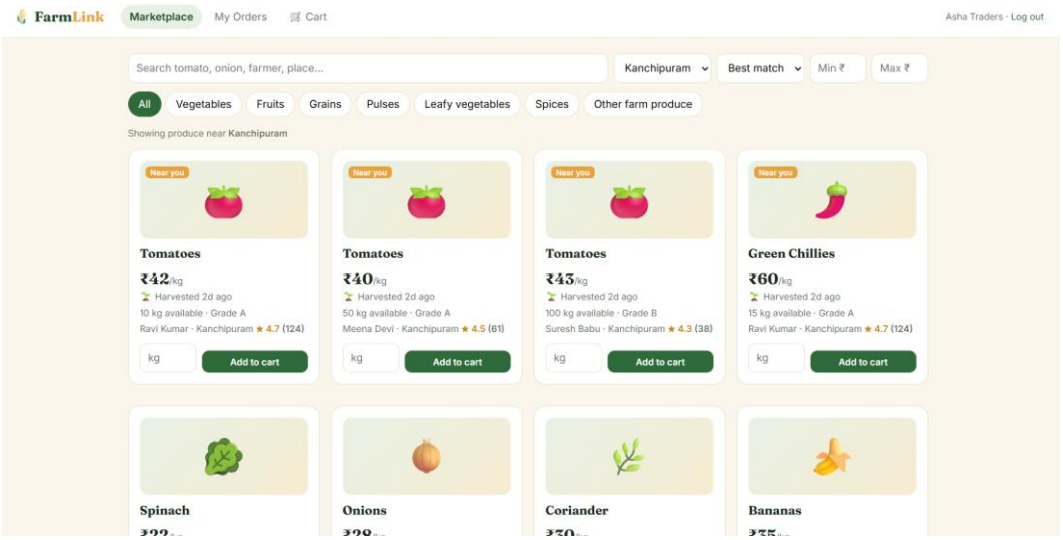


Figure 5.4: Buyer Marketplace

The order interface presents each order with its order number, status, ordered items, quantities, unit prices and line totals.

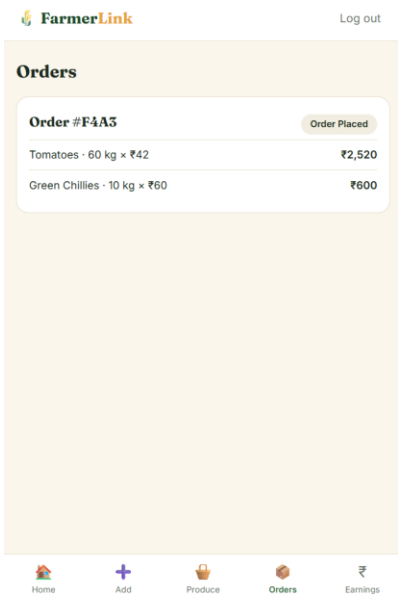


Figure 5.5: Order Summary and Status

5.1.6 Smart Batch Allocation

When a buyer places an order, the allocation engine identifies eligible inventory using a freshness-first, FIFO-like approach, with farmer rating applied as a tie-breaker when harvest dates are equal. For example, if a buyer requests 100 kg of Grade A Tomato and the Collection Center holds Batch 001 (Ramar, 60 kg), Batch 002 (Karthik, 40 kg) and Batch 003 (Mani, 50 kg), the system allocates 60 kg from Batch 001 and 40 kg from Batch 002, leaving Batch 003 untouched.

In the example, the order for 100 kg of Grade A tomato cannot be met from Batch 001 alone, which holds 60 kg. The engine therefore takes the whole of Batch 001 and then 40 kg from Batch 002, which is the next batch in priority. Batch 001 is consequently fully sold, with a sold quantity of 60 kg and no remaining quantity, Batch 002 is likewise fully sold, with a sold quantity of 40 kg and no remaining quantity, and Batch 003, holding 50 kg, is not touched. Had the order been for 50 kg, the whole requirement would have been taken from Batch 001, which would then show a sold quantity of 50 kg and a remaining quantity of 10 kg, and would remain active. The order is satisfied without the buyer having to name any farmer and each farmer's batch is updated by exactly the quantity taken from it.

The selection of Batches 001 and 002 in preference to Batch 003 reflects the freshness-first rule; in this example the earlier harvest times of Batches 001 and 002 give them priority. If, hypothetically, all three batches had the same harvest date, the farmer ratings would decide the order. With ratings of 4.8 for Karthik, 4.6 for Ramar and 4.2 for Mani, the batch of Karthik would be taken first, followed by that of Ramar. This hypothetical case is given only to show the operation of the tie-breaker and is not drawn from recorded data.

Following allocation, the Collection Center receives a digital picking instruction identifying the selected batches, ensuring that the correct physical produce is picked for dispatch.

5.1.7 Farmer Payment and Ledger

The payment system is designed to separate farmer earnings into an advance component and a final settlement component. The farmer ledger maintains the financial state of individual batches, distinguishing between quantities that are sold and eligible for final settlement and quantities that remain in inventory and are therefore pending. This prevents an entire batch from being incorrectly marked as sold when only a portion has been purchased.

Separating the advance from the final settlement allows the ledger to show a farmer's financial position at any point without ambiguity. When produce is accepted and the first tranche is paid, the advance appears as an entry against the batch. When the produce is sold and delivered, the final settlement appears as a second entry against the same batch. The amount not yet settled remains pending. At each moment the farmer can therefore distinguish between money already received, money that has become payable and money that depends on future sales.

The treatment of partial sales illustrates the value of this arrangement. Suppose that the baseline value of a 60 kg batch is ₹10,000, so that the advance is ₹7,000 and ₹3,000 remains. If 40 kg of the batch is sold, then, assuming for illustration that the baseline value is distributed evenly over the quantity, two thirds of the remaining amount, ₹2,000, relates to the sold quantity and becomes eligible for final settlement, whereas ₹1,000 relates to the 20 kg still in inventory and stays pending. The calculation

is given only to show how the ledger separates sold and unsold quantities. It prevents the whole batch from being treated as sold when only a part has been purchased.

The corresponding interface displays total earnings, current-month earnings, pending earnings and a history of completed orders, with earnings appearing once orders are delivered.

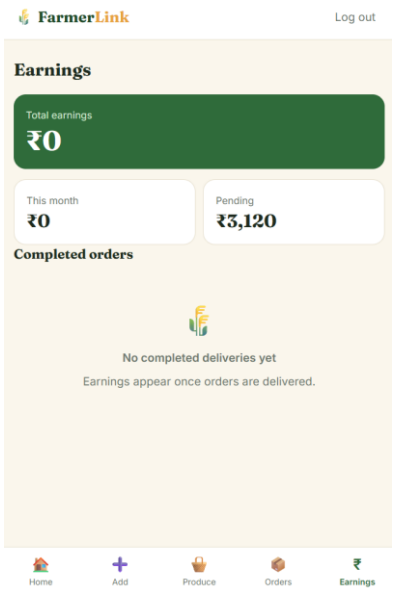


Figure 5.6: Farmer Earnings and Payment Ledger

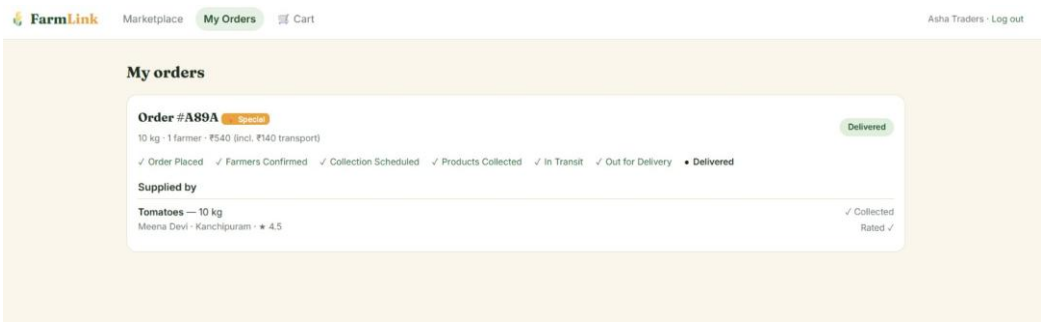


Figure 5.7 Order Tracking

5.1.8 Rating and Performance

The rating system maintains a farmer rating out of five based on buyer feedback regarding quality, grade accuracy, freshness and condition upon arrival. The proposed weighted moving-average formula is given below:

$$\textit{New Rating} = (\textit{Previous Rating} \times 0.7) + (\textit{Latest Order Rating} \times 0.3)$$

For example, if the previous rating is 4.5 and the latest order rating is 4.8, the new rating is calculated as (4.5 × 0.7) + (4.8 × 0.3) = 4.59. This method allows recent performance to influence the rating while retaining historical performance.

The weights of 0.7 and 0.3 define the balance between history and recent performance. Because the previous rating carries the greater weight, a single order cannot change the rating abruptly, and one unusually poor or unusually good delivery has a limited effect. Because the latest order still carries a weight of 0.3, sustained changes in performance are reflected within a small number of orders. Continuing the example, if the next order is rated 4.0, the rating moves from 4.59 to (4.59 × 0.7) + (4.0 × 0.3) = 4.413, or approximately 4.41, showing a modest decline rather than a sharp fall.

The rating is deliberately used as a secondary factor. Freshness remains the primary criterion of allocation, and the rating is consulted only when harvest dates are equal. The rating also appears in the marketplace listings, where it gives buyers an indication of the supplier's record. The combination provides a reason for farmers to maintain quality, grade accuracy, freshness and condition on arrival, without exposing perishable produce to the risk of being left unsold in favour of a better-rated but newer batch.

5.1.9 Collection Center Outwarding

Once an order is allocated, the Collection Center follows the outward workflow:

Buyer Order → Allocation Result → Picking Instructions → Scan/Confirm Batch → Load Produce → Dispatch

This workflow provides an organized mechanism for moving the correct physical batches toward fulfilment.

5.2 DISCUSSION

The FarmLink design focuses on solving the traceability and operational problems associated with aggregated agricultural inventory. A major advantage of the system is the use of digital batches. Instead of treating all produce at a Collection Center

as anonymous inventory, FarmLink maintains the relationship between the farmer, batch, inventory quantity, order allocation, sold quantity and payment.

The partial batch mechanism is particularly important because commercial orders may not exactly match individual farmer contributions. By maintaining sold and remaining quantities separately, the system preserves inventory accuracy.

The freshness-first allocation mechanism addresses the perishable nature of agricultural produce. Older batches receive priority, reducing the possibility of inventory aging. When freshness is equal, farmer rating acts as a secondary criterion.

The use of a simple, rule-based mechanism has the advantage of transparency. A farmer whose produce was not selected can be told that an older batch was selected first, and a Collection Center can verify the result by inspecting the harvest dates. The rating tie-breaker is similarly easy to explain. The mechanism does not rely on statistical estimation, and its results can be reproduced by anyone who has the batch records.

The two-tranche payment mechanism also addresses farmer liquidity. An initial advance can be provided after produce is accepted, while the remaining settlement is linked to the actual sale and fulfilment process. The system therefore provides benefits to all three participants, as summarized below.

Farmers

- Commercial market access
- Transparent batch tracking
- Advance payment and final settlement visibility
- Digital financial ledger
- Performance-based reputation
- Future harvest visibility

Collection Centers

- Digital inwarding and inventory control
- Batch traceability
- Structured picking and dispatch management

- Reduced manual record keeping

Buyers

- Aggregated supply and standardized grades
- Inventory visibility and transparent pricing
- Traceable sourcing
- Predictable fulfilment

These expected benefits are explicitly identified in the FarmLink design.

The two-tranche payment mechanism and the ledger together address the lack of payment visibility that was identified in the existing system. The first tranche reduces the delay between supply and payment, the second tranche links the remainder to actual sale and fulfilment, and the ledger shows the status of each batch. The farmer is therefore informed of what has been received and what is still awaited, and can relate each amount to the produce it was derived from.

It should be noted that this report documents the design of FarmLink and the implemented workflows and interfaces of the first phase. The benefits described are those that follow from the design. A study of the platform in extended operation with farmers, Collection Centers and buyers, and the associated measurements, would be a natural subsequent step and lies beyond the scope of the present phase.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

FarmLink – Agricultural Marketplace Platform provides a digitally traceable agricultural supply-chain model connecting smallholder farmers, Collection Centers and commercial buyers.

The system addresses the limitations of fragmented agricultural supply by introducing Collection Centers as physical aggregation and control points. Produce received at the Collection Center is weighed, graded and converted into a unique digital batch.

The most important feature of the platform is its batch-level traceability. Every physical quantity of produce remains digitally associated with the farmer who supplied it. This enables the system to track inventory, partial sales, order allocation and farmer payments accurately.

FarmLink further introduces freshness-first batch allocation, rating-based tie-breaking, two-tranche farmer settlement, digital farmer ledgers, quality-based ratings and transparent buyer pricing. The central relationship of the system can be represented as:

Farmer → Batch → Inventory → Order Allocation → Sold Quantity → Payment

This ensures that aggregated agricultural produce can be supplied to commercial buyers while maintaining visibility of individual farmer contributions and earnings. Thus, FarmLink provides a foundation for a transparent, organized and digitally traceable agricultural marketplace.

The work reported here is the first phase of the project and has concentrated on establishing the core workflow of the platform: the farmer-to-Collection-Center-to-buyer flow, together with batch management, allocation, payment tracking and ratings. The implemented interfaces demonstrate that this workflow can be expressed in a role-based web application built with React, Vite, Tailwind CSS, Node.js, Express, PostgreSQL and Prisma, with JWT-based authentication and Git and GitHub for

version control. The extensions described under Future Work are intended to build on this core once it has been validated.

6.2 FUTURE WORK

The project identifies several features that can be introduced once the core MVP workflow is stable.

Each of the following items is a possible extension and is not a part of the current implementation. They are listed approximately in the order in which they build on the existing workflow.

1. QR and Barcode Integration

Automated QR/barcode scanning can be implemented for faster batch identification and physical inventory tracking.

At present, batch identity is conveyed by the batch ID and the crate tag. Scanning would allow the Collection Center to identify a batch by reading the code on a crate, which should reduce manual entry at inwarding, picking and dispatch.

2. Advanced Logistics Optimization

Future versions can optimize routes and transportation for collecting and delivering aggregated produce.

Route optimization would address the coordination of vehicles for collecting produce from farmers and delivering to buyers, building on the allocation results that already identify which batches are to be dispatched.

3. Sophisticated Dynamic Pricing

Freshness-based pricing can be developed further using configurable pricing formulas.

Freshness-based pricing is part of the platform concept. Its further development would allow the way in which price responds to the age of produce to be defined through configurable formulas rather than fixed rules.

4. Payment Gateway Integration

Digital payment gateways can be integrated for automated farmer and buyer transactions.

Integration with payment gateways would allow the advances and final settlements calculated by the platform to be paid, and buyer payments to be received, electronically, so that the ledger entries correspond to processed transactions.

5. Predictive Analytics

Future harvest information can be used to estimate upcoming regional supply.

The upcoming harvest information already included in the farmer module could be aggregated to give an estimate of the supply that a region is likely to offer in the coming period.

6. Demand Forecasting

Historical order information can be analyzed to predict future buyer demand.

Order records accumulated over time form the basis for estimating the demand of buyers, which could then be compared with expected supply so that Collection Centers can prepare for periods of high or low activity.

7. Notifications

The platform can provide notifications for orders, payments, batch updates and fulfilment events.

Notifications would inform participants of events such as an order being placed, a batch being updated, a payment being made or an order being delivered, so that they need not check the application repeatedly.

8. Advanced Analytics

Dashboards can provide insights into inventory, farmer performance, sales, demand and regional supply.

Analytical dashboards would summarize the data already held by the platform, such as inventory levels, sales, farmer performance and demand, to support decisions by Collection Center managers and administrators.

9. Expanded Logistics Management

Future versions can provide more sophisticated transportation and dispatch management.

10. Large-Scale Deployment

The platform can be expanded to additional Collection Centers, regions, crops, farmers and commercial buyers.

Because batches and their quantities are independent of the place at which they are stored, expansion to further Collection Centers, regions and crops is largely a matter of adding data and operating units to the same design, and it is expected to follow validation of the core workflow.

The development strategy explicitly prioritizes completing and validating the core workflow before adding advanced logistics, dynamic pricing, QR scanning, payment gateways, predictive analytics and other advanced capabilities.

This sequence reflects a deliberate engineering decision. Features such as payment gateways, scanning and logistics optimization depend on a reliable core of batches, allocation and settlement, and introducing them earlier would add complexity before that core had been established.

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Abstract: Many farmers struggle to sell their produce in larger markets and make satisfactory prices, due to multiple intermediaries in the supply chain and lower proportion of the final price that goes to the farmer. At the same time, commercial buyers have difficulties acquiring a steady supply of produce with satisfactory quality from individual farmers. This paper presents FarmLink, a digital agricultural marketplace that brings together farmers, Collection Centers (CCs) and commercial buyers on a single platform. Produce delivered to CC is received, weighed and graded, and a digital record of a batch of produce with a unique identifier is created, with the ability to trace it back to an individual farmer, thus enabling per-farmer traceability even after being mixed with other batches. The platform allows for partial batch sales to be made, with the unsold amount going back into inventory. The platform facilitates smart allocation of produce to buyer orders, with prioritization based on freshness, and farmer rating as a tie-breaker in the case of similar harvest dates. A digital farmer ledger is maintained of advances, pending amounts and final settlements due to farmers. The platform is implemented in React with Vite and Tailwind CSS for the frontend, Node.js and Express for the backend, PostgreSQL with Prisma ORM for the database, and uses JWT-based role authentication for security. By connecting the digital marketplace with the physical produce, FarmLink creates transparency and traceability of agricultural products from farm to table.

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FarmLink: A Digital Agricultural Marketplace for Traceable Produce Aggregation, Smart Allocation, and Farmer Settlement

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Abstract - *Many farmers struggle to sell their produce in larger markets and make satisfactory prices, due to multiple intermediaries in the supply chain and lower proportion of the final price that goes to the farmer. At the same time, commercial buyers have difficulties acquiring a steady supply of produce with satisfactory quality from individual farmers. This paper presents FarmLink, a digital agricultural marketplace that brings together farmers, Collection Centers (CCs) and commercial buyers on a single platform. Produce delivered to CC is received, weighed and graded, and a digital record of a batch of produce with a unique identifier is created, with the ability to trace it back to an individual farmer, thus enabling per-farmer traceability even after being mixed with other batches. The platform allows for partial batch sales to be made, with the unsold amount going back into inventory. The platform facilitates smart allocation of produce to buyer orders, with prioritization based on freshness, and farmer rating as a tie-breaker in the case of similar harvest dates. A digital farmer ledger is maintained of advances, pending amounts and final settlements due to farmers. The platform is implemented in React with Vite and Tailwind CSS for the frontend, Node.js and Express for the backend, PostgreSQL with Prisma ORM for the database, and uses JWT-based role authentication for security. By connecting the digital marketplace with the physical produce, FarmLink creates transparency and traceability of agricultural products from farm to table.*

Keywords— *Agricultural Marketplace, FarmLink, Collection Center, Batch Traceability, Smart Allocation, Farmer Ledger, Inventory Management, Digital Agriculture.*

I. INTRODUCTION

Agriculture is the lifeline of many people and vital component of the economy. Small and medium-scale farmers however struggle to directly engage with larger buyers as their produce must pass through multiple layers of intermediaries before it reaches the commercial market, which reduces farmer incomes and complicates the supply chain.

Commercial buyers requiring large volumes of produce must aggregate from many individual farmers as there are inconsistencies in quantity, quality, harvest time and availability. While physical aggregation resolves the volume problem, recording and tracking ownership and remaining inventory can become a challenge.

Digital tools can help address this issue by creating a shared space for farmer and buyer engagement. However, an agricultural marketplace cannot exist purely on the digital space, as the physical handling of the produce is equally important. The product must be received, weighed, graded, stored, tracked and ultimately dispatched to the final buyer. The digital system must therefore also record what occurs to the produce at the Collection Center.

FarmLink has been designed accordingly and aims to connect farmers, Collection Centers and commercial buyers via a single medium. The Collection Center serves as the intermediate link to the marketplace: as the produce is brought in by the farmer, the quantity and grade are recorded digitally and a digital batch is created that is still linked to the farmer, allowing for traceability even once aggregated. A further aspect of the system is inventory management: a batch of produce does not have to be sold as a whole. If only a portion is bought, the amount sold is removed from the inventory while the remainder remains available. Orders are matched to batches based on freshness, with farmer rating used if the harvest dates are similar, with a digital ledger maintaining farmer payment information including advances, pending and settled payments. The objective of FarmLink is therefore not simply to create an online marketplace, but provide a connected system for produce aggregation, traceability, inventory management, order fulfilment and farmer settlement.

A. Objectives

The objectives of the FarmLink project are to:

- Provide a digital platform connecting farmers and commercial buyers via Collection Centers.
- Record and maintain produce received at the Collection Centers.
- Create unique digital batches for traceability.
- Track the quantity supplied by an individual farmer even after physical aggregation.
- Allow partial sales and maintain the remaining quantity in a certain batch.
- Help buyers find produce according to crop, quantity, grade and delivery location.
- Allocate available produce considering freshness and farmer rating.
- Maintain a digital ledger for payments to farmers and transaction history.
- Improve visibility and coordination between farmers, CCs and buyers.

B. Existing System

In the conventional agricultural supply chain, farmers often rely on middlemen or local market channels for selling their produce with no visibility over who their produce is reaching or how their crop is priced and distributed. Small

individual supply quantities also prevent them from bidding for bulk orders.

Digital agricultural marketplaces allow farmers to list produce and buyers to place orders, and to a certain extent, improve farming market access. However, just a marketplace is insufficient to solve the physical aggregation problem - after farmers supply their produce to a common location, the system should maintain information on the produce's sources (how much each farmer supplied in terms of quantity and grade) as well as its current status. Handling of partial orders is a limitation too - if a buyer places an order which is lower than the quantity supplied by a farmer, the remaining quantity should be tracked, and with no batch level inventory, it becomes challenging to accurately record this, especially if the buyer's requirement is fulfilled with combine produce of multiple farmers.

C. Proposed System

FarmLink proposes a three-sided agricultural marketplace linking Farmers, Collection Centers (CCs) and Commercial Buyers. Farmers supply produce to a Collection Center where it is received, weighed, inspected, graded and recorded, and a unique batch is created for the accepted produce containing details like the supplying farmer, crop, grade, quantity and other related transaction information thus enabling continuous connection between the physical produce and its sources. Commercial buyers may browse marketplace inventory and filter produce according to their requirements. Once an order is placed, the system identifies suitable batches for fulfillment, prioritizing freshness so that older suitable produce is allocated first; if there are multiple suitable batches with similar harvest dates, farmer rating is used as a tie-breaker. The system also allows partial batch sales, say a 60 kg batch being used to fulfill a 40 kg order, with just 40 kg marked as sold while the remaining 20 kg continues to be visible in the inventory thus enabling the batch to be traced until its lifecycle ends. A digital farmer ledger maintains advance payments, pending earnings, final settlements and transaction history, giving farmers insight into the financials of their supplied produce. By combining these functions, FarmLink seeks to make agricultural procurement organized while improving traceability, inventory visibility, order fulfillment and payment transparency.

II. LITERATURE REVIEW

Digital technologies have been increasingly used to enhance the marketing and distribution of agricultural produce. Prior work has explored electronic marketplaces, digital platforms, farmer participation, Farmer Producer Organisations (FPOs), and digital supply chains, generally finding that online platforms enhance market access, information sharing, and coordination while highlighting the complexities of adoption, trust, infrastructure, awareness, and the limited scale of individual farmers.

Studies looking at digital trade and electronic marketplaces provide the background to this work. Das et al. [1] explored eNAM and digital agricultural trade, finding that digital trading systems enhance the access to markets but do not address the physical aggregation and batch level tracking of produce on large scale. Argade et al.

[2] studied electronic marketplaces and stressed the importance of information sharing between participants, indicating that quantity, quality, availability, and delivery information must be maintained and shared to facilitate coordination. Raghavendra et al. [3] used survey based analysis to identify factors influencing farmer participation in electronic marketing, while Agnihotri and Bhattacharya [4] found that trust and perceived usefulness strongly influence digital marketplace adoption implying that a farmer facing platform needs to be simple and transparent to gain reliance.

A group of related studies look at collective participation. Shroff [5] discussed how farmer collectives enhance the market access for marginal farmers, and Kumar and Kumar [6] looked at digitalization within the FPO business ecosystem. These are relevant to FarmLink as Collection Centers offer a structure through which the produce of multiple farmers may be aggregated while retaining the farmer level records.

Recent work has focused more directly on digital agricultural platforms. Devika et al. [7] proposed a mobile platform for direct market access, Mishra et al. [8] studied digital e-marketplaces and identified digital literacy as a key adoption factor, Nandhini et al. [9] studied digital marketing in agriculture and how it may enhance market reach, and Khan [10] explored agricultural e-commerce and the infrastructure challenges affecting implementation. A further set of studies look at auction and supply-chain models: Dave and Sajja [11] proposed an e-auction model for FPOs, Ravi Kumar et al. [12] showed that access to information influences participant participation in chilli e-markets, and Gummagolmath and Lakshmi [13] discussed FPO federation models addressing the constraints due to scale. Kumar et al. [14] identified market related constraints affecting FPO performance, Pendyala et al.

[15] stressed the importance of awareness for successful adoption of e-commerce among FPOs, Singh et al. [16] showed how better coordination improves the performance of e-agricultural supply chains, and Kumar and Parshant [17] presented ARGOMARKET, a web based agricultural marketplace demonstrating the feasibility of direct digital trading between farmers and buyers.

Taken together, these works show that digital platforms can enhance the market access of agricultural produce and improve communication between producers and buyers, but most concentrate on a single element online trading, farmer participation, FPO coordination, or market information, giving less attention to the physical handling of produce after its arrival at an aggregation point. Batch level traceability when produce from multiple farmers is combined, flexible handling of partial sales, freshness aware order fulfilment, and integrated farmer settlement remains largely unaddressed as an integrated system. This is where FarmLink focuses its attention: instead of treating a farmer's produce as a mere online listing, FarmLink creates a digital batch when the produce is received and accepted at the Collection Center, retains that batch linked to the farmer through storage, sale, allocation, and dispatch, supports partial batch sales while retaining the ability to track remaining quantity, apply freshness first allocation with farmer rating as a tie-breaker and maintain a farmer level ledger for payments tracking, combining market access with traceability, inventory control, order fulfilment and settlement management in one platform.

III. METHODOLOGY

FarmLink is a role-based digital marketplace designed to connect farmers, Collection Centers (CCs), and commercial buyers. The methodology focuses on maintaining a connection between the physical movement of agricultural produce and its corresponding digital records; instead of viewing produce only as an online listing, the system creates and manages digital batches from the point at which produce is received at the Collection Center.

A. System Workflow

The FarmLink workflow begins when a farmer supplies produce to a Collection Center. The received produce is weighed, inspected, and graded by the Collection Center, and once accepted, a unique digital batch is created containing details such as the farmer, crop, grade, quantity, and relevant timestamps. The accepted produce is then added to the Collection Center's inventory; since each batch remains associated with its supplying farmer, the system can determine the source and remaining quantity of the produce even after aggregation. When a buyer places an order, the system checks available inventory and selects suitable batches using a freshness-first approach, preferring produce with earlier harvest dates and using farmer rating as a tie-breaker when harvest dates are similar. After allocation, the Collection Center receives the required picking information and prepares the produce for dispatch; once the order is fulfilled, the corresponding quantities are updated and the farmer's payment records are adjusted, creating a continuous flow from produce receipt to final settlement.

The system follows a layered web architecture, illustrated in Fig. 3.1. The frontend is developed using React with Vite and Tailwind CSS, providing separate role-based interfaces for farmers, Collection Centers, and buyers. The backend is implemented using Node.js and Express, handling business logic, authentication, inventory operations, order processing, allocation, and payment related records, with Prisma ORM used to communicate with the PostgreSQL database. JWT-based authentication secures role-based access across all layers.

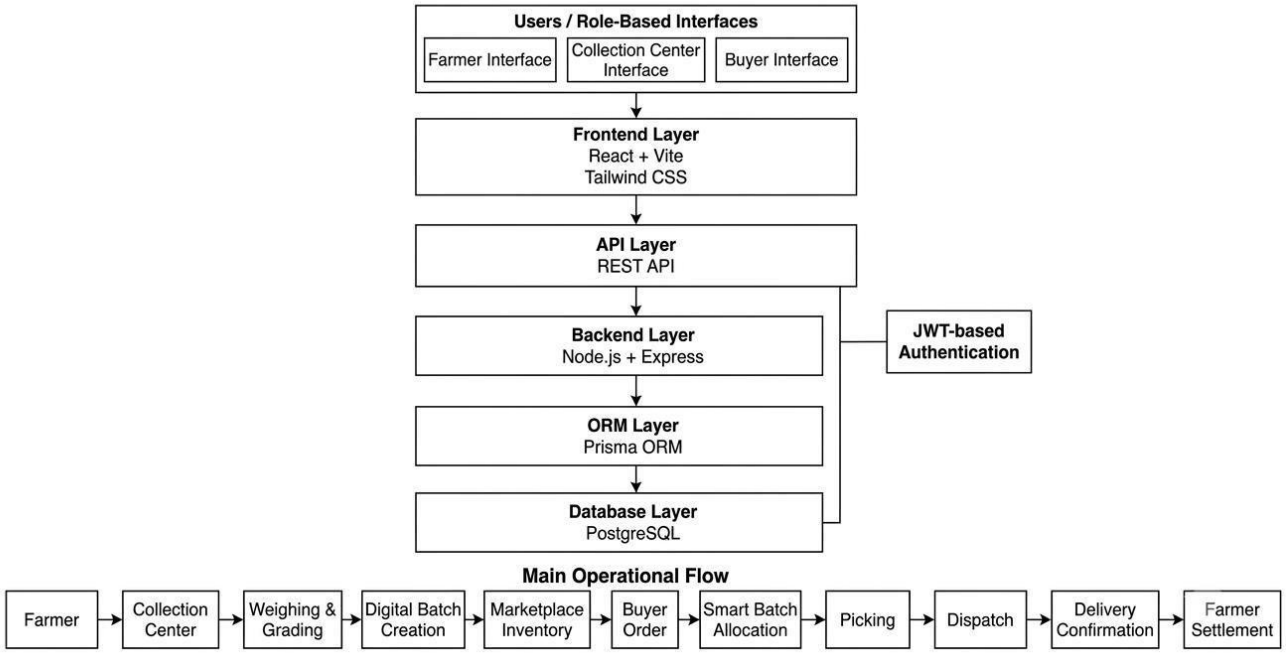


Fig. 3.1. System Architecture of FarmLink

As shown in the architecture, user requests flow from the role-based frontend, through a REST API to the Express backend, which applies business rules before reading from or writing to the PostgreSQL database through Prisma ORM. JWT-based authentication sits alongside the backend and API layers, verifying the role of each request so that farmers, Collection Center operators and buyers only access the functions relevant to their role. The lower part of the figure traces the main operational flow of produce - from the farmer, through Collection Center inwarding, weighing and grading, digital batch creation, marketplace inventory, buyer ordering, smart batch allocation, picking, dispatch, delivery confirmation and farmer settlement - showing how each physical stage corresponds to an update in the digital record.

The operational sequence that underlies this architecture is represented in more detail as an activity flow, shown in Fig. 3.2. The activity begins with farmer produce submission and proceeds through Collection Center inwarding, weighing, grading, batch creation and inventory addition, prior to moving into buyer order placement, batch allocation, picking, dispatch and settlement.

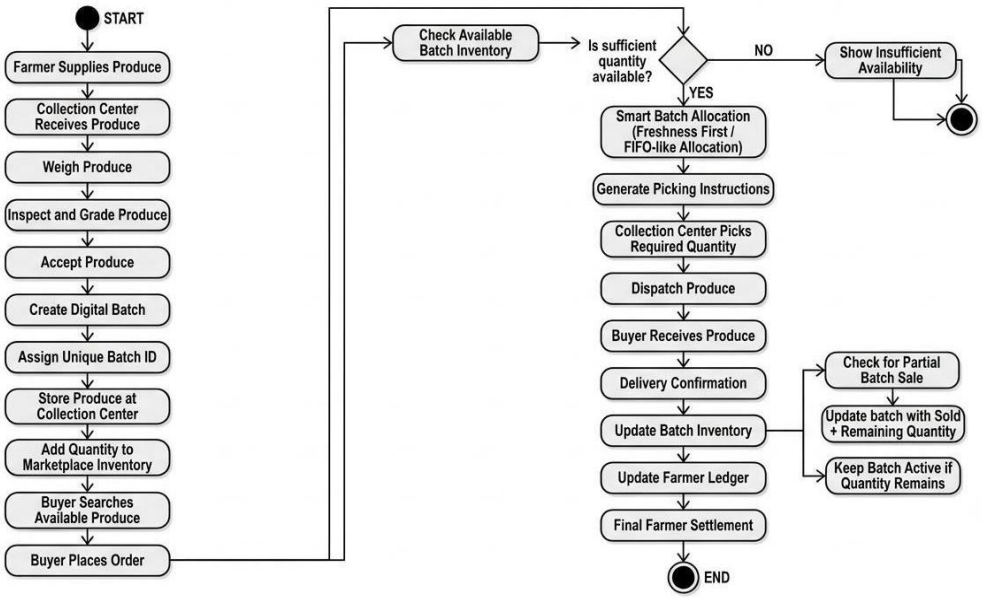


Fig. 3.2. Activity Diagram of FarmLink

As the activity diagram shows, once produce has been accepted and stored, it becomes available in marketplace inventory for buyers to search and order. When a buyer places an order, the system checks available batch inventory for sufficient quantity; if the available quantity is insufficient, the buyer is informed accordingly, and if sufficient produce is available, the system proceeds to smart batch allocation using the freshness-first rule described earlier. The Collection Center then receives picking instructions, dispatches the produce and the buyer's receipt and delivery confirmation trigger an update to batch inventory. If only part of a batch was required, the system checks for a partial sale, updates the batch with the sold and remaining quantities and keeps the batch active while quantity remains. This inventory update in turn updates the farmer ledger and completes the final farmer settlement for that transaction.

The high-level exchange of information between the main participants in this workflow is represented as a Level 0 data flow diagram, shown in Fig. 3.3.

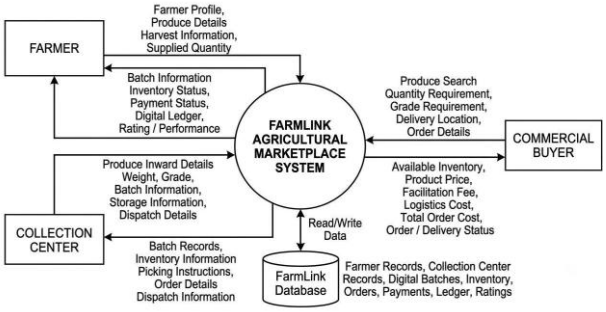


Fig. 3.3. Level 0 Data Flow Diagram of FarmLink

At this level, the Farmer, Collection Center and Commercial Buyer interact with the central FarmLink Agricultural Marketplace System, which in turn reads from and writes to the FarmLink database. Farmers supply profile, produce and harvest information and receive batch information, inventory status, payment status, ledger and rating updates in return. Collection Centers submit produce inward details, weight, grade, batch and storage information and receive batch records, inventory information, picking instructions and order and dispatch details. Commercial buyers submit produce search criteria, quantity, grade and delivery requirements and receive available inventory, pricing, facilitation and logistics costs, total order cost and order or delivery status. This high-level view sets the boundary of the system and the information exchanged across it, without detailing the internal processes that handle this exchange.

These internal processes are expanded in the Level 1 data flow diagram shown in Fig. 3.4.

The Level 1 data flow diagram divides the FarmLink system into five main processes: Farmer Management, Collection Center and Batch Management, Buyer and Order Management, Smart Allocation and Fulfilment and Payment, Ledger and Rating Management. The Farmer Management process records farmer profile and harvest information and returns dashboard, batch history, earnings, ledger and rating data. The Collection Center and Batch Management process receives inwarding, weight, grade and acceptance details from the Collection Center and produce supply information from farmers, and maintains batch and inventory records in the FarmLink database. The Buyer and Order Management process accepts search criteria and orders from buyers and returns available produce, pricing and order status, passing confirmed order and quantity requirements to the Smart Allocation and Fulfilment process. This process applies the freshness-first allocation rule described earlier, using farmer rating as a tie-breaker when harvest dates are equal, and returns picking instructions and dispatch information to the Collection Center while updating order fulfilment status for the buyer. Finally, the Payment, Ledger and Rating Management process updates the farmer ledger with advances, settlements and pending earnings once an order is fulfilled, and updates farmer rating and performance based on buyer feedback. All five processes read from and write to the shared FarmLink database, which stores farmer, Collection Center, batch, inventory, order, payment, ledger and rating data.

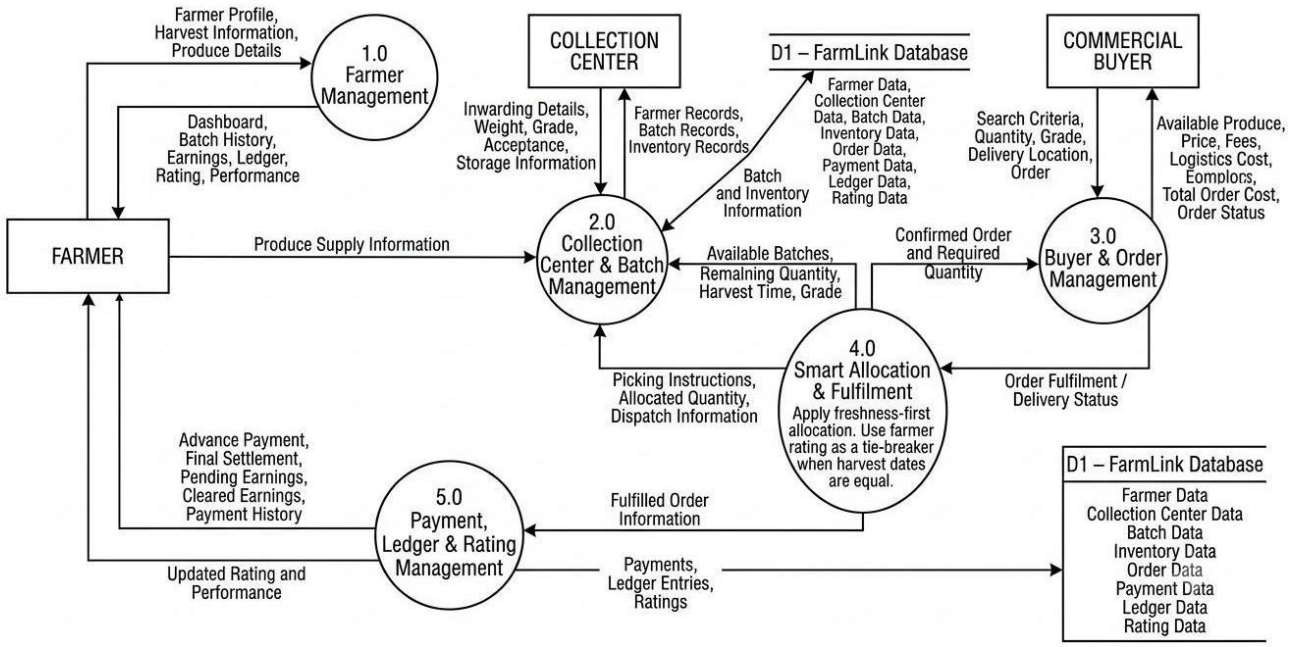


Fig. 3.4. Level 1 Data Flow Diagram of FarmLink

B. Batch-Level Traceability

The first crucial component of the methodology is digitally tagged batches with associated supply farmer details and current status. An accepted quantity of produce is allocated a unique batch ID that contains information regarding the supplying farmer, crop, grade, quantity, harvest or inwarding details, and the current status. For example, a batch of tomato can be represented as:

Batch ID: BATCH-TOM-RAMAR-001

Farmer: Ramar | Crop: Tomato | Grade: A | Quantity: 60 kg

If a buyer buys 40 kg tomato from this batch, quantity is updated to 40 kg against ramar's account with 20 kg left in the system as the available stock, without affecting the batch traceability to the supplying farmer throughout the product life cycle.

C. Smart Allocation and Settlement

Farmlink makes fresh produce selection prioritized based on the harvest date, ensuring old produce does not get stored where young produce gets dispatched first. In cases where the harvest date is the same or close, farmer rating gets prioritized and where quantities are limited, a smart selection can be made using multiple batch quantities to suffice an order against a single farmer category.

The methodology of settlements is broken down into two tranches with the first tranche representing an upfront payment made upon receiving, weighing, grading, and accepting the produce. The second tranche represents the counterpart of the settlement that is released either after selling the produce or completing the order. The proportion between the two can be configured based on business rules configured within the platform.

The farmer's ledger gets updated to record all transactions for reference, including cleared earnings, pending settlements, advances, final settlements, and history of transactions, providing a clear link between the produce supplied and the realized settlements in monetary terms.

The farmer's rating gets updated based on a buyer's feedback on the quality, grade specification accuracy, freshness, and general condition of the received produce. A weighted rating system is also applied to avoid spikes in ratings brought about by a single buyer's input, thus,

New Rating = (Previous Rating × 0.7) + (Latest Order Rating × 0.3) this ensures ratings change gradually as new buyers continue to interact with the platform. It can be noted that the methodology combined batching while tracing to the farmer, inventory management, freshness prioritized allocation, order fulfillment, and digital settlement into a digital agricultural marketplace.

IV. PROJECT DESCRIPTION

A. System Description

FarmLink is implemented as a role-based agricultural marketplace which coordinates the activities of farmers, Collection Centers and commercial buyers, connecting the digital marketplace with the physical handling of agricultural produce so that produce can be tracked from the time it is received at a Collection Center until it is sold and dispatched. The workflow begins with the farmer supplying produce to a Collection Center, which records its quantity, checks its quality, assigns a grade and creates a digital batch linked to the supplying farmer. Batches are maintained as inventory, and when a buyer places an order, the system identifies suitable available batches based on crop, grade, quantity and freshness, giving preference to earlier harvested batches and using farmer rating when available batches have similar harvest dates. Partial sales are supported throughout, so that only the sold portion of a batch is deducted while the remainder continues to be available under the same batch record, preventing loss of traceability across multiple transactions. Payments are handled through the two-stage advance-and-settlement process described earlier, with all transactions reflected in the farmer's digital ledger.

B. Modules

FarmLink is organized into five major modules that manage the different activities involved in agricultural trading.

1) Farmer Management Module

This module maintains the profile and supply-related information of each farmer. Farmers can view their dashboard, supplied quantities, batch history, ratings, earnings and transaction details. The module also maintains the farmer's digital ledger, separating cleared earnings, pending earnings, advances and final settlements, giving farmers a clear view of the money associated with the produce they have supplied.

2) Collection Center and Batch Management Module

This module handles the physical inwarding and digital recording of produce. When produce reaches the Collection Center, it is weighed, inspected, graded and accepted before a batch is created; each batch receives a unique batch ID and contains details such as the farmer, crop, grade, quantity, and relevant timestamps. The module also maintains inventory and tracks the remaining quantity after sales, and the inventory can be viewed through a structured hierarchy such as Crop → Grade → Farmer → Batch → Remaining Quantity → Shelf-Life Status, making it easier for operators to locate and manage specific quantities.

3) Buyer and Order Management Module

This module gives commercial buyers access to available agricultural produce. Buyers can search and filter produce by crop, quantity, minimum grade and delivery location. Before confirming an order, the buyer can view the product price along with applicable platform facilitation and logistics costs; the system then records the order and passes the requirement to the allocation process.

4) Smart Allocation and Fulfilment Module

This module determines which available batches should be used to fulfil a buyer's order, applying freshness-first allocation and using farmer rating as a tie-breaker when harvest dates are similar. For example, if a buyer requires 100 kg of Grade A tomatoes and the available inventory consists of 60 kg from one farmer, 40 kg from another and 50 kg from a third, the system can allocate the first two batches to fulfil the order while leaving the third batch untouched. After allocation, the Collection Center receives the required picking information, and the selected produce is picked, prepared and dispatched to the buyer.

5) Payment, Ledger and Rating Module

This module manages farmer settlements and performance information. Payments are recorded in two stages — an initial advance after accepted inwarding and a final settlement after the corresponding produce is sold or fulfilled — and a transaction history is maintained so that the relationship between supplied batches, sales and farmer earnings can be tracked. The rating component updates a farmer's rating based on buyer feedback on quality, grade accuracy, freshness and condition of the produce, using the weighted update described in Section III-C so that the rating changes gradually rather than being dominated by a single recent order.

V. IMPLEMENTATION AND RESULT DISCUSSION

A. Implementation Results

FarmLink has been implemented as a web-based agricultural marketplace with distinct interfaces for farmers, Collection Centers, and commercial buyers. The implementation emphasizes maintaining consistent records across produce handling, inventory, orders, allocation, and farmer settlements.

1) Farmer Management

Farmer interface allows a view of their profile info, provided produce, batch history, ratings, and payment details, and their dashboard shows cleared and pending earnings. The digital ledger maintains the transaction history of the farmer's provided produce, allowing a view of their advances, final settlements, and pending amounts.

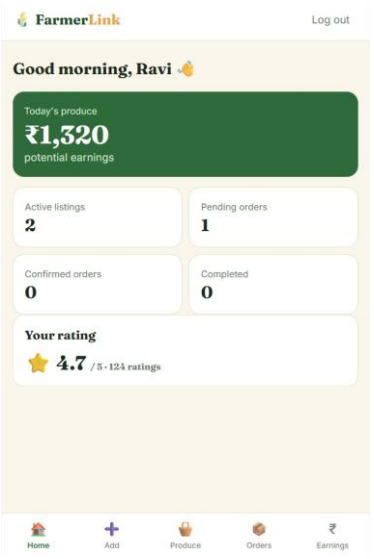


Fig. 5.1. Farmer Dashboard

2) Collection Center Inwarding

The Collection Center interface is used when produce is physically received from farmers. The operator records the crop, quantity, detail, and grade after inspection, and after acceptance, a corresponding digital record is made, which becomes the starting point of the produce tracking throughout the other marketplace workflow.

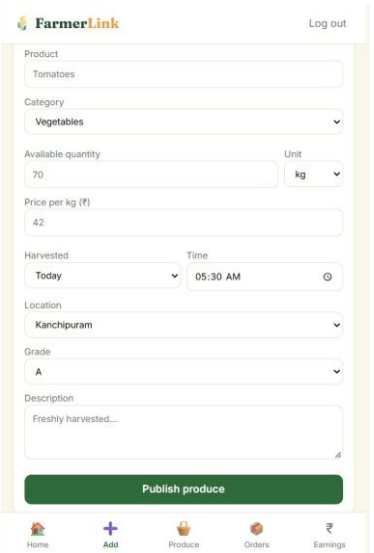


Fig. 5.2. Produce Entry Interface

3) Digital Batch and Inventory Management

After the inwarding, the accepted produce is assigned a unique batch ID with information about the supplying farmer, crop, grade, quantity, and any relevant timestamps. The batch is added to the Collection Center inventory and since it is linked to the farmer, the system can identify the source of the produce even if produce from multiple farmers have been aggregated.

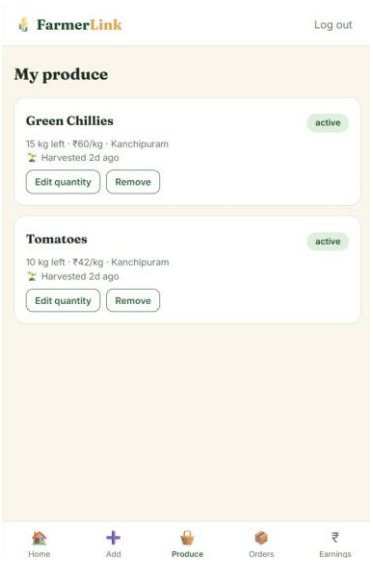


Fig. 5.3. Produce Listing and Inventory Management

4) Partial Batch Sales

FarmLink allows a partial sale of an available batch (e.g. 60 kg batch and order of 40 kg) by recording the 40 kg as sold and retaining the 20 kg as available inventory. The batch is active until its available quantity reaches 0, thus avoiding unnecessary duplicate records but still tracing the original produce.

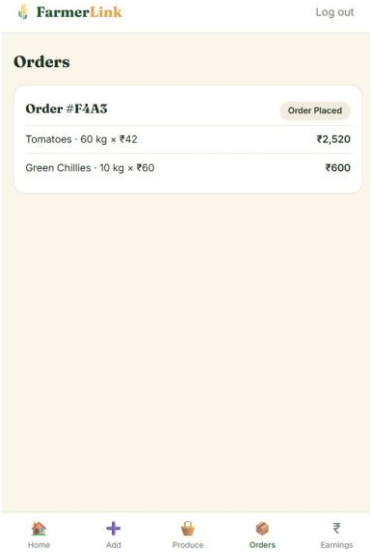


Fig. 5.4. Farmer Order View with Quantity Drawn from Listings

5) Buyer Marketplace

The buyer interface shows available agricultural produce, allows searching according to their requirements, with filters including crop, quantity, minimum grade, and delivery location. The order summary breaks down the product price, platform facilitation fee, logistics cost, and total order cost before an order is confirmed.

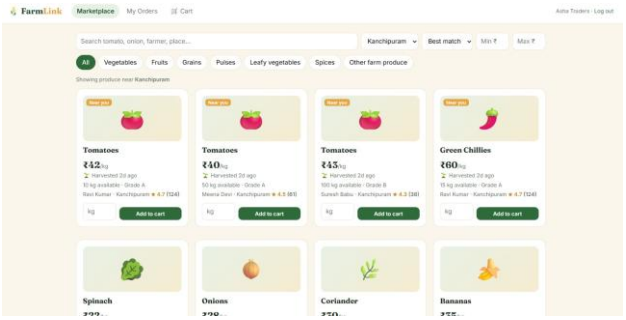


Fig. 5.5. Buyer Marketplace

6) Smart Batch Allocation

After placing an order, the system identifies suitable batches from the available inventory and prioritizes batches with earlier harvest dates. For cases with two or more batches with similar harvest dates, farmer rating is used as a tie-breaking factor, and the system can combine quantities from multiple batches when a batch is not enough to satisfy the complete order.

[Insert Smart Batch Allocation Screenshot Here]

7) Farmer Payment, Ledger, and Rating

The payment process records farmer earnings in two stages as described in Section III-C, and the farmer ledger provides a consolidated view of cleared, pending amounts, advances, and final settlements. Buyer feedback is used to maintain the farmer's performance rating, considering factors such as produce quality, grade accuracy, freshness, and condition upon arrival with the weighted update allowing the latest feedback to impact the rating without discarding their previous performance record.

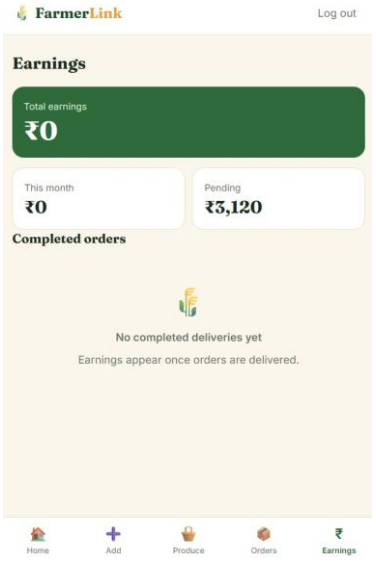


Fig. 5.6. Farmer Earnings and Ledger View

8) Collection Center Outwarding

Once the required batches are allocated, the Collection Center prepares the produce for outwarding and the system provides the information needed to identify the selected batches and quantities. After the produce is dispatched, the corresponding inventory quantities are updated, thus com-

pleting the movement of produce from Collection Center inventory to the buyer.

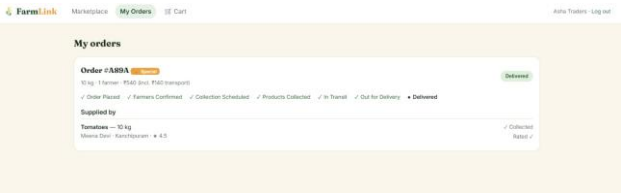


Fig. 5.7. Order Tracking and Delivery Status

B. Discussion

The implementation showcases how FarmLink merges marketplace operations with physical produce management. A key design choice is the maintenance of digital records at the batch level rather than just on overall inventory to maintain farmer-level traceability even if produce from multiple farmers are aggregated at a Collection Center - directly addressing the gap identified in Section II, where existing digital marketplaces were found to be giving comparatively little attention to physical aggregation and batch-level tracking.

The partial-sale mechanism allows for more flexible inventory, since a batch does not need to be sold in one transaction and its remaining quantity can still participate in future orders. The freshness-first allocation approach is a practical way of managing the agricultural inventory, and the farmer rating mechanism provides an additional decision-making factor when batches have similar freshness. The two-stage payment process and digital ledger further enhance transparency by linking produce supply directly to farmer earnings. Since these features are part of the system's design rather than the subject of a separate field trial, they are presented here as implemented but not yet experimentally validated at scale; no performance figures, adoption statistics, or user-satisfaction measures are reported since none have been collected. Overall, the implemented modules work together to provide a structured flow from farmer supply, through Collection Center handling, digital batch creation, marketplace ordering, batch allocation, and fulfilment, to farmer settlement.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

FarmLink introduces a digital agricultural marketplace to facilitate transactions between farmers, Collection Centers, and commercial buyers, emphasizing not only online buying and selling but also physical handling and tracking of agricultural produce. With digital batches, every quantity of produce maintains its association with its supplying farmer throughout the inventory and order process, and features such as partial batch sales, freshness-first allocation, farmer ratings, and Collection Center-based inventory management offer a more organized approach to agricultural fulfilment. The two-stage payment process and digital farmer ledger make it easier to track advances, pending earnings, and final settlements, and buyers are provided with structured information on produce availability, grade, quantity, pricing, and delivery-related costs. Overall, FarmLink aims to reduce the gap between farmers and commercial buyers by combining marketplace access, physical aggregation, batch-level traceability, inventory management, order fulfilment, and payment tracking within one system, providing a stronger foundation for transparent and coordinated agricultural transactions.

B. Future Work

The current system can be enhanced in several ways to improve its usability and scale. Future versions of FarmLink can include:

- QR and barcode-based batch tracking for faster identification of physical produce.
- Mobile applications for farmers, Collection Centers, and buyers.
- Real-time logistics tracking for better visibility during transportation.
- Demand and price prediction to support farmer and buyer planning decisions.
- Automated freshness monitoring using sensors or IoT devices for stored produce.
- Multi-language support to make the platform accessible to a wider range of users.
- Advanced analytics dashboards for monitoring farmer performance, inventory movement, sales, and demand.
- Scalable cloud deployment to support a larger number of farmers, Collection Centers, and buyers across regions.

These improvements can further enhance FarmLink as a scalable digital platform for agricultural supply-chain management.

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FarmLink: A Digital Agricultural Marketplace for Traceable Produce Aggregation, Smart Allocation, and Farmer Settlement

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***Abstract*— *Many farmers struggle to sell their produce in larger markets and make satisfactory prices, due to multiple intermediaries in the supply chain and lower proportion of the final price that goes to the farmer. At the same time, commercial buyers have difficulties acquiring a steady supply of produce with satisfactory quality from individual farmers. This paper presents FarmLink, a digital agricultural marketplace that brings together farmers, Collection Centers (CCs) and commercial buyers on a single platform. Produce delivered to CC is received, weighed and graded, and a digital record of a batch of produce with a unique identifier is created, with the ability to trace it back to an individual farmer, thus enabling per-farmer traceability even after being mixed with other batches. The platform allows for partial batch sales to be made, with the unsold amount going back into inventory. The platform facilitates smart allocation of produce to buyer orders, with prioritization based on freshness, and farmer rating as a tie-breaker in the case of similar harvest dates. A digital farmer ledger is maintained of advances, pending amounts and final settlements due to farmers. The platform is implemented in React with Vite and Tailwind CSS for the frontend, Node.js and Express for the backend, PostgreSQL with Prisma ORM for the database, and uses JWT-based role authentication for security. By connecting the digital marketplace with the physical produce, FarmLink creates transparency and traceability of agricultural products from farm to table.*

***Keywords*—** *Agricultural Marketplace, FarmLink, Collection Center, Batch Traceability, Smart Allocation, Farmer Ledger, Inventory Management, Digital Agriculture.*

I. INTRODUCTION

Agriculture is the lifeline of many people and vital component of the economy. Small and medium-scale farmers however struggle to directly engage with larger buyers as their produce must pass through multiple layers of intermediaries before it reaches the commercial market, which reduces farmer incomes and complicates the supply chain.

Commercial buyers requiring large volumes of produce must aggregate from many individual farmers as there are inconsistencies in quantity, quality, harvest time and availability. While physical aggregation resolves the volume problem, recording and tracking ownership and remaining inventory can become a challenge.

Digital tools can help address this issue by creating a shared space for farmer and buyer engagement. However, an agricultural marketplace cannot exist purely on the digital space, as the physical handling of the produce is equally important. The product must be received, weighed, graded, stored, tracked and ultimately dispatched to the final buyer. The digital system must therefore also record what occurs to the produce at the Collection Center.

FarmLink has been designed accordingly and aims to connect farmers, Collection Centers and commercial buyers via a single medium. The Collection Center serves as the intermediate link to the marketplace: as the produce is brought in by the farmer, the quantity and grade are recorded digitally and a digital batch is created that is still linked to the farmer, allowing for traceability even once aggregated. A further aspect of the system is inventory management: a batch of produce does not have to be sold as a whole. If only a portion is bought, the amount sold is removed from the inventory while the remainder remains available. Orders are matched to batches based on freshness, with farmer rating used if the harvest dates are similar, with a digital ledger maintaining farmer payment information including advances, pending and settled payments. The objective of FarmLink is therefore not simply to create an online marketplace, but provide a connected system for produce aggregation, traceability, inventory management, order fulfilment and farmer settlement.

A. Objectives

The objectives of the FarmLink project are to:

- Provide a digital platform connecting farmers and commercial buyers via Collection Centers.
- Record and maintain produce received at the Collection Centers.
- Create unique digital batches for traceability.
- Track the quantity supplied by an individual farmer even after physical aggregation.
- Allow partial sales and maintain the remaining quantity in a certain batch.

- Help buyers find produce according to crop, quantity, grade and delivery location.
- Allocate available produce considering freshness and farmer rating.
- Maintain a digital ledger for payments to farmers and transaction history.
- Improve visibility and coordination between farmers, CCs and buyers.

B. Existing System

In the conventional agricultural supply chain, farmers often rely on middlemen or local market channels for selling their produce with no visibility over who their produce is reaching or how their crop is priced and distributed. Small individual supply quantities also prevent them from bidding for bulk orders.

Digital agricultural marketplaces allow farmers to list produce and buyers to place orders, and to a certain extent, improve farming market access. However, just a marketplace is insufficient to solve the physical aggregation problem - after farmers supply their produce to a common location, the system should maintain information on the produce's sources (how much each farmer supplied in terms of quantity and grade) as well as its current status. Handling of partial orders is a limitation too - if a buyer places an order which is lower than the quantity supplied by a farmer, the remaining quantity should be tracked, and with no batch level inventory, it becomes challenging to accurately record this, especially if the buyer's requirement is fulfilled with combine produce of multiple farmers.

C. Proposed System

FarmLink proposes a three-sided agricultural marketplace linking Farmers, Collection Centers (CCs) and Commercial Buyers. Farmers supply produce to a Collection Center where it is received, weighed, inspected, graded and recorded, and a unique batch is created for the accepted produce containing details like the supplying farmer, crop, grade, quantity and other related transaction information thus enabling continuous connection between the physical produce and its sources. Commercial buyers may browse marketplace inventory and filter produce according to their requirements. Once an order is placed, the system identifies suitable batches for fulfillment, prioritizing freshness so that older suitable produce is allocated first; if there are multiple suitable batches with similar harvest dates, farmer rating is used as a tie-breaker. The system also allows partial batch sales, say a 60 kg batch being used to fulfill a 40 kg order, with just 40 kg marked as sold while the remaining 20 kg continues to be visible in the inventory thus enabling the batch to be traced until its lifecycle ends. A digital farmer ledger maintains advance payments, pending earnings, final settlements and transaction history, giving farmers insight into the financials of their supplied produce. By combining these functions, FarmLink seeks to make agricultural procurement organized while improving traceability, inventory visibility, order fulfillment and payment transparency.

II. LITERATURE REVIEW

Digital technologies have been increasingly used to enhance the marketing and distribution of agricultural produce. Prior work has explored electronic marketplaces, digital platforms, farmer participation, Farmer Producer Organisations (FPOs), and digital supply chains, generally finding that online platforms enhance market access, information sharing, and coordination while highlighting the complexities of adoption, trust, infrastructure, awareness, and the limited scale of individual farmers.

Studies looking at digital trade and electronic marketplaces provide the background to this work. Das et al. [1] explored eNAM and digital agricultural trade, finding that digital trading systems enhance the access to markets but do not address the physical aggregation and batch level tracking of produce on large scale. Argade et al. [2] studied electronic marketplaces and stressed the importance of information sharing between participants, indicating that quantity, quality, availability, and delivery information must be maintained and shared to facilitate coordination. Raghavendra et al. [3] used survey based analysis to identify factors influencing farmer participation in electronic marketing, while Agnihotri and Bhattacharya [4] found that trust and perceived usefulness strongly influence digital marketplace adoption implying that a farmer facing platform needs to be simple and transparent to gain reliance.

A group of related studies look at collective participation. Shroff [5] discussed how farmer collectives enhance the market access for marginal farmers, and Kumar and Kumar [6] looked at digitalization within the FPO business ecosystem. These are relevant to FarmLink as Collection Centers offer a structure through which the produce of multiple farmers may be aggregated while retaining the farmer level records.

Recent work has focused more directly on digital agricultural platforms. Devika et al. [7] proposed a mobile platform for direct market access, Mishra et al. [8] studied digital e-marketplaces and identified digital literacy as a key adoption factor, Nandhini et al. [9] studied digital marketing in agriculture and how it may enhance market reach, and Khan [10] explored agricultural e-commerce and the infrastructure challenges affecting implementation. A further set of studies look at auction and supply-chain models: Dave and Sajja [11] proposed an e-auction model for FPOs, Ravi Kumar et al. [12] showed that access to information influences participant participation in chilli e-markets, and Gummagolmath and Lakshmi [13] discussed FPO federation models addressing the constraints due to scale. Kumar et al. [14] identified market related constraints affecting FPO performance, Pendyala et al. [15] stressed the importance of awareness for successful adoption of e-commerce among FPOs, Singh et al. [16] showed how better coordination improves the performance of e-agricultural supply chains, and Kumar and Parshant [17] presented ARGOMARKET, a web based agricultural marketplace demonstrating the feasibility of direct digital trading between farmers and buyers.

Taken together, these works show that digital platforms can enhance the market access of agricultural produce and

improve communication between producers and buyers, but most concentrate on a single element online trading, farmer participation, FPO coordination, or market information, giving less attention to the physical handling of produce after its arrival at an aggregation point. Batch level traceability when produce from multiple farmers is combined, flexible handling of partial sales, freshness aware order fulfilment, and integrated farmer settlement remains largely unaddressed as an integrated system. This is where FarmLink focuses its attention: instead of treating a farmer's produce as a mere online listing, FarmLink creates a digital batch when the produce is received and accepted at the Collection Center, retains that batch linked to the farmer through storage, sale, allocation, and dispatch, supports partial batch sales while retaining the ability to track remaining quantity, apply freshness first allocation with farmer rating as a tie-breaker and maintain a farmer level ledger for payments tracking, combining market access with traceability, inventory control, order fulfilment and settlement management in one platform.

III. METHODOLOGY

FarmLink is a role-based digital marketplace designed to connect farmers, Collection Centers (CCs), and commercial buyers. The methodology focuses on maintaining a connection between the physical movement of agricultural produce and its corresponding digital records; instead of viewing produce only as an online listing, the system creates and manages digital batches from the point at which produce is received at the Collection Center.

A. System Workflow

The FarmLink workflow begins when a farmer supplies produce to a Collection Center. The received produce is weighed, inspected, and graded by the Collection Center, and once accepted, a unique digital batch is created containing details such as the farmer, crop, grade, quantity, and relevant timestamps. The accepted produce is then added to the Collection Center's inventory; since each batch remains associated with its supplying farmer, the system can determine the source and remaining quantity of the produce even after aggregation. When a buyer places an order, the system checks available inventory and selects suitable batches using a freshness-first approach, preferring produce with earlier harvest dates and using farmer rating as a tie-breaker when harvest dates are similar. After allocation, the Collection Center receives the required picking information and prepares the produce for dispatch; once the order is fulfilled, the corresponding quantities are updated and the farmer's payment records are adjusted, creating a continuous flow from produce receipt to final settlement.

The system follows a layered web architecture, illustrated in Fig. 3.1. The frontend is developed using React with Vite and Tailwind CSS, providing separate role-based interfaces for farmers, Collection Centers, and buyers. The backend is implemented using Node.js and Express, handling business logic, authentication, inventory operations, order processing, allocation, and payment related records, with Prisma ORM used to communicate with the PostgreSQL database. JWT-based authentication secures role-based access across all layers.

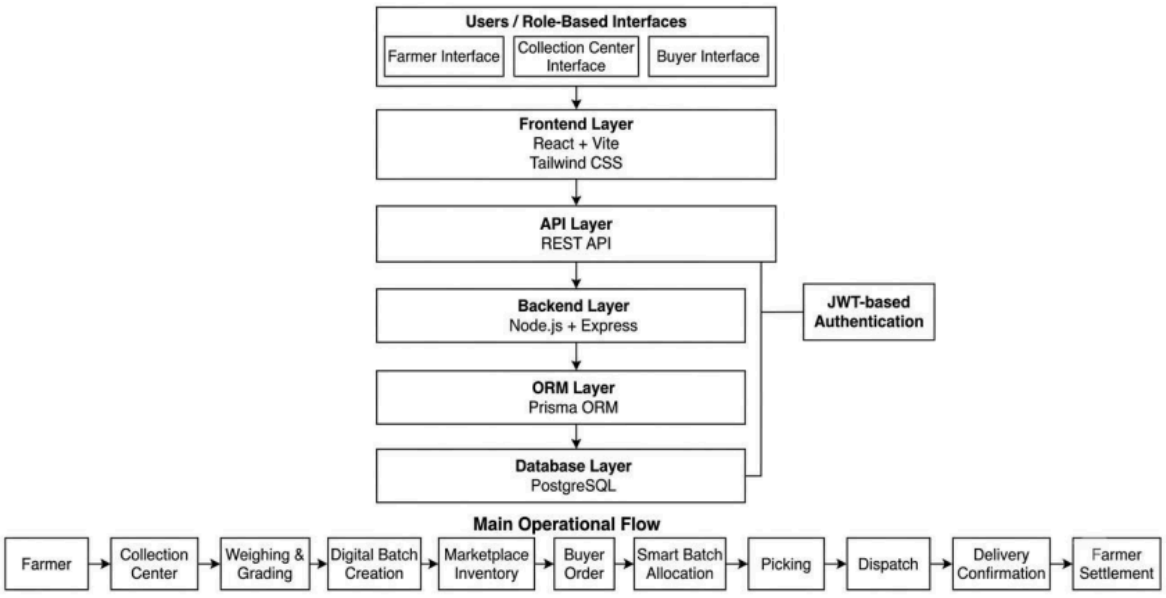


Figure 3.1: System Architecture of FarmLink

Fig. 3.1. System Architecture of FarmLink

As shown in the architecture, user requests flow from the role-based frontend, through a REST API to the Express backend, which applies business rules before reading from or writing to the PostgreSQL database through Prisma ORM. JWT-based authentication sits alongside the backend and API layers, verifying the role of each request so that farmers, Collection Center operators and buyers only access the functions relevant to their role. The lower

part of the figure traces the main operational flow of produce - from the farmer, through Collection Center inwarding, weighing and grading, digital batch creation, marketplace inventory, buyer ordering, smart batch allocation, picking, dispatch, delivery confirmation and farmer settlement - showing how each physical stage corresponds to an update in the digital record.

The operational sequence that underlies this architecture is represented in more detail as an activity flow, shown in Fig. 3.2. The activity begins with farmer produce submission and proceeds through Collection Center inwarding, weighing, grading, batch creation and inventory addition, prior to moving into buyer order placement, batch allocation, picking, dispatch and settlement.

Figure 3.2: Activity Diagram of FarmLink

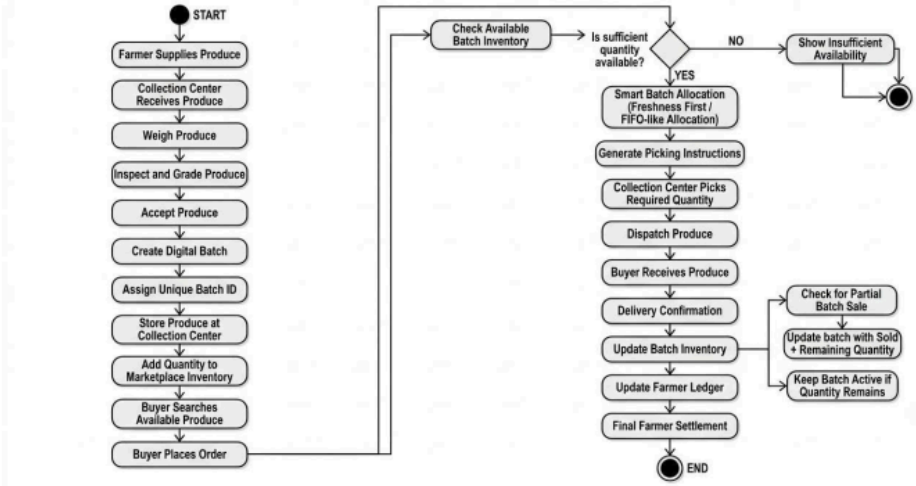


Fig. 3.2. Activity Diagram of FarmLink

As the activity diagram shows, once produce has been accepted and stored, it becomes available in marketplace inventory for buyers to search and order. When a buyer places an order, the system checks available batch inventory for sufficient quantity; if the available quantity is insufficient, the buyer is informed accordingly, and if sufficient produce is available, the system proceeds to smart batch allocation using the freshness-first rule described earlier. The Collection Center then receives picking instructions, dispatches the produce and the buyer's receipt and delivery confirmation trigger an update to batch inventory. If only part of a batch was required, the system checks for a partial sale, updates the batch with the sold and remaining quantities and keeps the batch active while quantity remains. This inventory update in turn updates the farmer ledger and completes the final farmer settlement for that transaction.

The high-level exchange of information between the main participants in this workflow is represented as a Level 0 data flow diagram, shown in Fig. 3.3.

Figure 3.3: Level 0 Data Flow Diagram of FarmLink

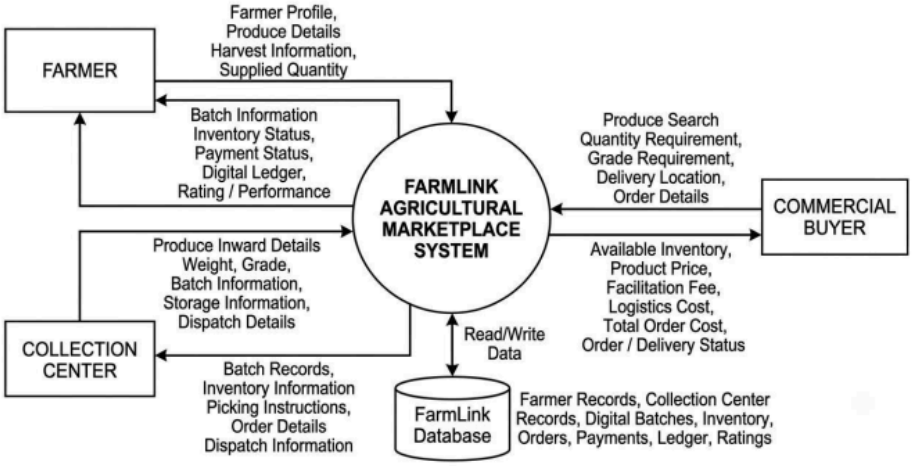


Fig. 3.3. Level 0 Data Flow Diagram of FarmLink

At this level, the Farmer, Collection Center and Commercial Buyer interact with the central FarmLink Agricultural Marketplace System, which in turn reads from and writes to the FarmLink database. Farmers supply profile, produce and harvest information and receive batch information, inventory status, payment status, ledger and rating updates in return. Collection Centers submit produce inward details, weight, grade, batch and storage information and receive batch records, inventory information, picking instructions and order and dispatch details. Commercial buyers submit produce search criteria, quantity, grade and delivery requirements and receive available inventory, pricing, facilitation and logistics costs, total order cost and order or delivery status. This high-level view sets the boundary of the system and the information exchanged across it, without detailing the internal processes that handle this exchange.

These internal processes are expanded in the Level 1 data flow diagram shown in Fig. 3.4.

Figure 3.4: Level 1 Data Flow Diagram of FarmLink

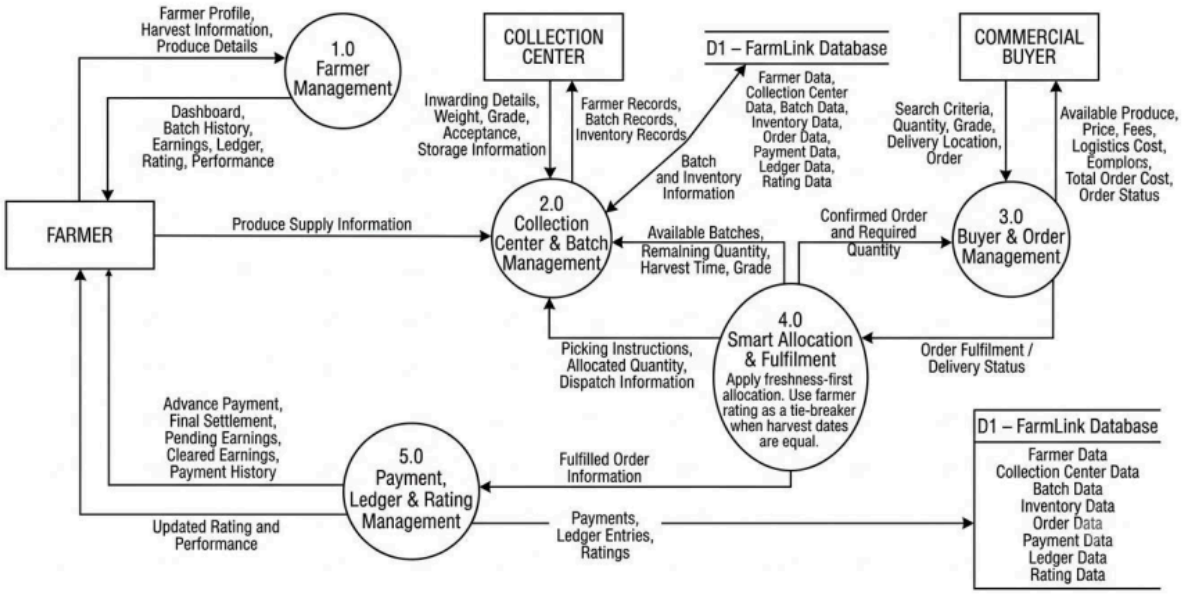


Figure 3.4: Level 1 Data Flow Diagram of FarmLink

Fig. 3.4. Level 1 Data Flow Diagram of FarmLink

The Level 1 data flow diagram divides the FarmLink system into five main processes: Farmer Management, Collection Center and Batch Management, Buyer and Order Management, Smart Allocation and Fulfilment and Payment, Ledger and Rating Management. The Farmer Management process records farmer profile and harvest information and returns dashboard, batch history, earnings, ledger and rating data. The Collection Center and Batch Management process receives inwarding, weight, grade and acceptance details from the Collection Center and produce supply information from farmers, and maintains batch and inventory records in the FarmLink database. The Buyer and Order Management process accepts search criteria and orders from buyers and returns available produce, pricing and order status, passing confirmed order and quantity requirements to the Smart Allocation and Fulfilment process. This process applies the freshness-first allocation rule described earlier, using farmer rating as a tie-breaker when harvest dates are equal, and returns picking instructions and dispatch information to the Collection Center while updating order fulfilment status for the buyer. Finally, the Payment, Ledger and Rating Management process updates the farmer ledger with advances, settlements and pending earnings once an order is fulfilled, and updates farmer rating and performance based on buyer feedback. All five processes read from and write to the shared FarmLink database, which stores farmer, Collection Center, batch, inventory, order, payment, ledger and rating data.

B. Batch-Level Traceability

The first crucial component of the methodology is digitally tagged batches with associated supply farmer details and current status. An accepted quantity of produce is allocated a unique batch ID that contains information regarding the supplying farmer, crop, grade, quantity, harvest or inwarding details, and the current status. For example, a batch of tomato can be represented as:

Batch ID: BATCH-TOM-RAMAR-001
Farmer: Ramar | Crop: Tomato | Grade: A | Quantity: 60 kg

If a buyer buys 40 kg tomato from this batch, quantity is updated to 40 kg against ramar's account with 20 kg left in the system as the available stock, without affecting the batch traceability to the supplying farmer throughout the product life cycle.

C. Smart Allocation and Settlement

Farmlink makes fresh produce selection prioritized based on the harvest date, ensuring old produce does not get

stored where young produce gets dispatched first. In cases where the harvest date is the same or close, farmer rating gets prioritized and where quantities are limited, a smart selection can be made using multiple batch quantities to suffice an order against a single farmer category.

The methodology of settlements is broken down into two tranches with the first tranche representing an upfront payment made upon receiving, weighing, grading, and accepting the produce. The second tranche represents the counterpart of the settlement that is released either after selling the produce or completing the order. The proportion between the two can be configured based on business rules configured within the platform.

The farmer's ledger gets updated to record all transactions for reference, including cleared earnings, pending settlements, advances, final settlements, and history of transactions, providing a clear link between the produce supplied and the realized settlements in monetary terms.

The farmer's rating gets updated based on a buyer's feedback on the quality, grade specification accuracy, freshness, and general condition of the received produce. A weighted rating system is also applied to avoid spikes in ratings brought about by a single buyer's input, thus,

$$\text{New Rating} = (\text{Previous Rating} \times 0.7) + (\text{Latest Order Rating} \times 0.3)$$

this ensures ratings change gradually as new buyers continue to interact with the platform. It can be noted that the methodology combined batching while tracing to the farmer, inventory management, freshness prioritized allocation, order fulfillment, and digital settlement into a digital agricultural marketplace.

IV. PROJECT DESCRIPTION

A. System Description

FarmLink is implemented as a role-based agricultural marketplace which coordinates the activities of farmers, Collection Centers and commercial buyers, connecting the digital marketplace with the physical handling of agricultural produce so that produce can be tracked from the time it is received at a Collection Center until it is sold and dispatched. The workflow begins with the farmer supplying produce to a Collection Center, which records its quantity, checks its quality, assigns a grade and creates a digital batch linked to the supplying farmer. Batches are maintained as inventory, and when a buyer places an order, the system identifies suitable available batches based on crop, grade, quantity and freshness, giving preference to earlier harvested batches and using farmer rating when available batches have similar harvest dates. Partial sales are supported throughout, so that only the sold portion of a batch is deducted while the remainder continues to be available under the same batch record, preventing loss of traceability across multiple transactions. Payments are handled through the two-stage advance-and-settlement process described earlier, with all transactions reflected in the farmer's digital ledger.

B. Modules

FarmLink is organized into five major modules that manage the different activities involved in agricultural trading.

1) Farmer Management Module

This module maintains the profile and supply-related information of each farmer. Farmers can view their dashboard, supplied quantities, batch history, ratings, earnings and transaction details. The module also maintains the farmer's digital ledger, separating cleared earnings, pending earnings, advances and final settlements, giving farmers a clear view of the money associated with the produce they have supplied.

2) Collection Center and Batch Management Module

This module handles the physical inwarding and digital recording of produce. When produce reaches the Collection Center, it is weighed, inspected, graded and accepted before a batch is created; each batch receives a unique batch ID and contains details such as the farmer, crop, grade, quantity, and relevant timestamps. The module also maintains inventory and tracks the remaining quantity after sales, and the inventory can be viewed through a structured hierarchy such as Crop → Grade → Farmer → Batch → Remaining Quantity → Shelf-Life Status, making it easier for operators to locate and manage specific quantities.

3) Buyer and Order Management Module

This module gives commercial buyers access to available agricultural produce. Buyers can search and filter produce by crop, quantity, minimum grade and delivery location. Before confirming an order, the buyer can view the product price along with applicable platform facilitation and logistics costs; the system then records the order and passes the requirement to the allocation process.

4) Smart Allocation and Fulfilment Module

This module determines which available batches should be used to fulfil a buyer's order, applying freshness-first allocation and using farmer rating as a tie-breaker when harvest dates are similar. For example, if a buyer requires 100 kg of Grade A tomatoes and the available inventory consists of 60 kg from one farmer, 40 kg from another and 50 kg from a third, the system can allocate the first two batches to fulfil the order while leaving the third batch untouched. After allocation, the Collection Center receives the required picking information, and the selected produce is picked, prepared and dispatched to the buyer.

5) Payment, Ledger and Rating Module

This module manages farmer settlements and performance information. Payments are recorded in two stages — an initial advance after accepted inwarding and a final settlement after the corresponding produce is sold or fulfilled — and a transaction history is maintained so that the relationship between supplied batches, sales and farmer earnings can be tracked. The rating component updates a farmer's rating based on buyer feedback on quality, grade accuracy, freshness and condition of the produce, using the weighted update described in Section III-C so that the

rating changes gradually rather than being dominated by a single recent order.

V. IMPLEMENTATION AND RESULT DISCUSSION

A. Implementation Results

FarmLink has been implemented as a web-based agricultural marketplace with distinct interfaces for farmers, Collection Centers, and commercial buyers. The implementation emphasizes maintaining consistent records across produce handling, inventory, orders, allocation, and farmer settlements.

1) Farmer Management

Farmer interface allows a view of their profile info, provided produce, batch history, ratings, and payment details, and their dashboard shows cleared and pending earnings. The digital ledger maintains the transaction history of the farmer's provided produce, allowing a view of their advances, final settlements, and pending amounts.

[Insert FarmLink Farmer Dashboard Screenshot Here]

2) Collection Center Inwarding

The Collection Center interface is used when produce is physically received from farmers. The operator records the crop, quantity, detail, and grade after inspection, and after acceptance, a corresponding digital record is made, which becomes the starting point of the produce tracking throughout the other marketplace workflow.

[Insert Collection Center Inwarding Interface Screenshot Here]

3) Digital Batch and Inventory Management

After the inwarding, the accepted produce is assigned a unique batch ID with information about the supplying farmer, crop, grade, quantity, and any relevant timestamps. The batch is added to the Collection Center inventory and since it is linked to the farmer, the system can identify the source of the produce even if produce from multiple farmers have been aggregated.

[Insert Digital Batch Creation and Inventory Management Screenshots Here]

4) Partial Batch Sales

FarmLink allows a partial sale of an available batch (e.g. 60 kg batch and order of 40 kg) by recording the 40 kg as sold and retaining the 20 kg as available inventory. The batch is active until its available quantity reaches 0, thus avoiding unnecessary duplicate records but still tracing the original produce.

[Insert Partial Batch Sale Screenshot Here]

5) Buyer Marketplace

The buyer interface shows available agricultural produce, allows searching according to their requirements, with filters including crop, quantity, minimum grade, and delivery location. The order summary breaks down the product price, platform facilitation fee, logistics cost, and total order cost before an order is confirmed.

[Insert Buyer Marketplace and Order Summary Screenshots Here]

6) Smart Batch Allocation

After placing an order, the system identifies suitable batches from the available inventory and prioritizes batches with earlier harvest dates. For cases with two or more batches with similar harvest dates, farmer rating is used as a tie-breaking factor, and the system can combine quantities from multiple batches when a batch is not enough to satisfy the complete order.

[Insert Smart Batch Allocation Screenshot Here]

7) Farmer Payment, Ledger, and Rating

The payment process records farmer earnings in two stages as described in Section III-C, and the farmer ledger provides a consolidated view of cleared, pending amounts, advances, and final settlements. Buyer feedback is used to maintain the farmer's performance rating, considering factors such as produce quality, grade accuracy, freshness, and condition upon arrival with the weighted update allowing the latest feedback to impact the rating without discarding their previous performance record.

[Insert Farmer Payment, Ledger, and Rating Screenshots Here]

8) Collection Center Outwarding

Once the required batches are allocated, the Collection Center prepares the produce for outwarding and the system provides the information needed to identify the selected batches and quantities. After the produce is dispatched, the corresponding inventory quantities are updated, thus completing the movement of produce from Collection Center inventory to the buyer.

[Insert Collection Center Outwarding and Dispatch Screenshot Here]

B. Discussion

The implementation showcases how FarmLink merges marketplace operations with physical produce management. A key design choice is the maintenance of digital records at the batch level rather than just on overall inventory to maintain farmer-level traceability even if produce from multiple farmers are aggregated at a Collection Center - directly addressing the gap identified in Section II, where existing digital marketplaces were found to be giving comparatively little attention to physical aggregation and batch-level tracking.

The partial-sale mechanism allows for more flexible inventory, since a batch does not need to be sold in one transaction and its remaining quantity can still participate in future orders. The freshness-first allocation approach is a practical way of managing the agricultural inventory, and the farmer rating mechanism provides an additional decision-making factor when batches have similar freshness. The two-stage payment process and digital ledger further enhance transparency by linking produce supply directly to farmer earnings. Since these features are part of the system's design rather than the subject of a separate field trial, they are presented here as implemented but not yet experimentally validated at scale; no performance figures, adoption statistics, or user-satisfaction measures are reported since none have been

collected. Overall, the implemented modules work together to provide a structured flow from farmer supply, through Collection Center handling, digital batch creation, marketplace ordering, batch allocation, and fulfilment, to farmer settlement.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

FarmLink introduces a digital agricultural marketplace to facilitate transactions between farmers, Collection Centers, and commercial buyers, emphasizing not only online buying and selling but also physical handling and tracking of agricultural produce. With digital batches, every quantity of produce maintains its association with its supplying farmer throughout the inventory and order process, and features such as partial batch sales, freshness-first allocation, farmer ratings, and Collection Center-based inventory management offer a more organized approach to agricultural fulfilment. The two-stage payment process and digital farmer ledger make it easier to track advances, pending earnings, and final settlements, and buyers are provided with structured information on produce availability, grade, quantity, pricing, and delivery-related costs. Overall, FarmLink aims to reduce the gap between farmers and commercial buyers by combining marketplace access, physical aggregation, batch-level traceability, inventory management, order fulfilment, and payment tracking within one system, providing a stronger foundation for transparent and coordinated agricultural transactions.

B. Future Work

The current system can be enhanced in several ways to improve its usability and scale. Future versions of FarmLink can include:

- QR and barcode-based batch tracking for faster identification of physical produce.
- Mobile applications for farmers, Collection Centers, and buyers.
- Real-time logistics tracking for better visibility during transportation.
- Demand and price prediction to support farmer and buyer planning decisions.
- Automated freshness monitoring using sensors or IoT devices for stored produce.
- Multi-language support to make the platform accessible to a wider range of users.
- Advanced analytics dashboards for monitoring farmer performance, inventory movement, sales, and demand.
- Scalable cloud deployment to support a larger number of farmers, Collection Centers, and buyers across regions.

These improvements can further enhance FarmLink as a scalable digital platform for agricultural supply-chain management.

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